CITY OF COLORADO SPRINGS - EPC (CITY)
30 SOUTH NEVADA AVENUE
COLORADO SPRINGS, COLORADO 80901

TEMPERATURE CONTROL DRAWINGS

ADDITIONAL PROJECTS

(A) 03-29-08 TC3082 COLORADO SPGS CITY CAB 6TH FL VAV

(B) 09-08-08 TC3179 COLORADO SPGS CITY CAB RTU ACTUATORS

LONG

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"AS BUILT"

LONG
Denver 303-975-2100 Colorado Springs 719-592-0364 Wyoming 307-265-5997

CITY OF COLORADO SPRINGS - EPC (CITY)
COVER SHEET
30 SOUTH NEVADA AVENUE
COLORADO SPRINGS, COLORADO 80901

NO.	Revision:	Date:	By:
A	ENGINEERING REVIEW	10-12-07	BD
B	TC3082 AS BUILT	03-28-08	JWV
C	TC3082 AS BUILT	05-15-08	CGD
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E			
F			

Engineer: BURNS & MCDONNELL
Contractor: BURNS & MCDONNELL
Proj Mgr.: KERRY NOBLITT
App. Eng.: B. DHAKAL
Created: 08-22-07

Standard Template: N/A

JOB NUMBER TC2955
SHEET NO. 1

ABBREVIATIONS

TAG	DESCRIPTION	TAG	DESCRIPTION
AC	AIR COMPRESSOR	SD	SMOKE DETECTOR
AD	AIR DRYER	SG	STEAM GENERATOR
ADJ	ADJ	SNAFU	SITUATION NORMAL, ALL FOULED UP
AI	ANALOG INPUT	SPT	STATIC PRESSURE TRANSMITTER
ALM	ALARM	SS	START / STOP
AM	AMMETER	STS	STATUS
AO	ANALOG OUTPUT	SV	SOLENOID VALVE
AP	ALARM PANEL	SW	SWITCH
AQ	AIR QUALITY	T	THERMOSTAT
AR	ALARM RELAY	TB	TERMINAL BLOCK
AUX	AUXILIARY	TC	TIME CLOCK
BFP	BOILER FEEDWATER PUMP	TCP	CONTROL PANEL
CB	CIRCUIT BREAKER	TDR	TIME DELAY RELAY
CFP	CHEMICAL FEED PUMP	TK	PNEUMATIC THERMOSTAT
CHWP	CHILLED WATER PUMP	TR	TRANSFORMER
CHWR	CHILLED WATER RETURN	TS	TEMPERATURE SENSOR
CHWS	CHILLED WATER SUPPLY	TT	TEMPERATURE TRANSMITTER
CI	CONTACT INPUT	UH	UNIT HEATER
CO	CONTACT OUTPUT	UI	UNIVERSAL INPUT
COM	COMMON	V	VALVE
CRAC	COMPUTER ROOM AIR CONDITIONER	VA	VOLTAMPS
CRP	CONDENSER RETURN PUMP	VAC	VOLTS ALTERNATING CURRENT
CS	CURRENT SENSING SWITCH	VAV	VARIABLE AIR VOLUME
CT	CURRENT TRANSFORMER	VDC	VOLTS DIRECT CURRENT
CUH	CABINET UNIT HEATER	VFD	VARIABLE FREQUENCY DRIVE
CWP	CONDENSER WATER PUMP	VM	VOLTMETER
CWR	CONDENSER WATER RETURN		
CWS	CONDENSER WATER SUPPLY		
DMPR	DAMPER		
D.A.	DIRECT ACTING		
DA	DISCHARGE AIR		
DAD	DISCHARGE AIR DAMPER		
DAT	DISCHARGE AIR TEMPERATURE		
DDC	DIRECT DIGITAL CONTROL		
DHWP	DOMESTIC HOT WATER PUMP		
DHWR	DOMESTIC HOT WATER RECIRC.		
DHWS	DOMESTIC HOT WATER SUPPLY		
DI	DIGITAL INPUT		
DIS	DISABLE		
DM	DAMPER MOTOR		
DO	DIGITAL OUTPUT		
DPS	DIFFERENTIAL PRESS. SWITCH		
DPT	DIFFERENTIAL PRESS. TRANSMITTER		
DTC	DATA TERMINATION CABINET		
DWH	DOMESTIC WATER HEATER		
DX	DIRECT EXPANSION COOLING		
(E)	EXISTING		
EA	EXHAUST AIR		
EAD	EXHAUST AIR DAMPER		
EDH	ELECTRIC DUCT HEATER		
EF	EXHAUST FAN		
ENA	ENABLE		
EP	ELECTRIC/PNEUMATIC VALVE		
EPT	VOLTAGE TO PNEUMATIC TRANSDUCER		
ES	END SWITCH		
F	FUSE		
FAP	FIRE ALARM PANEL		
FCU	FAN COIL UNIT		
FE	FLOW ELEMENT		
FLS	FLOAT SWITCH		
FOP	FUEL OIL PUMP		
FS	FLOW SWITCH		
FT	FLOW TRANSMITTER		
FTP	FUEL OIL TRANSFER PUMP		
GA	GAUGE		
GND	GROUND		
L1	120V HOT		
H	HUMIDISTAT		
HT	HUMIDITY TRANSMITTER		
HWP	HEATING WATER PUMP		
HWR	HEATING WATER RETURN		
HWS	HEATING WATER SUPPLY		
HX	HEAT EXCHANGER		
KWM	KW METER		
LLSV	LIQUID LINE SOLENOID VALVE		
LS	LIMIT SWITCH		
M	MOTOR		
mA	MILLIAMPS		
MA	MIXED AIR		
MAG	MAGNETIC MOTOR STARTER		
MAT	MIXED AIR TEMPERATURE		
MAX	MAXIMUM		
MCC	MOTOR CONTROL CENTER		
MDM	MODEM		
MH	MAGNAHELIC		
L2	120V NEUTRAL		
N.C.	NORMALLY CLOSED		
N.O.	NORMALLY OPEN		
OA	OUTSIDE AIR		
OAD	OUTSIDE AIR DAMPER		
OAT	OUTSIDE AIR TEMPERATURE		
OL	OVERLOAD		
OS	OCCUPANCY SENSOR		
PB	PUSHBUTTON		
PC	PERSONAL COMPUTER		
PE	PNEUMATIC/ELECTRIC SWITCH		
PL	PILOT LIGHT		
POT	POTENTIOMETER		
PS	PRESSURE SWITCH		
PSI	POUNDS PER SQ. INCH		
PSIA	POUNDS PER SQ. INCH ABSOLUTE		
PSID	POUNDS PER SQ. INCH DIFF.		
PSIG	POUNDS PER SQ. INCH GAUGE		
R.A.	REVERSE ACTING		
RA	RETURN AIR		
RAD	RETURN AIR DAMPER		
RAT	RETURN AIR TEMPERATURE		
RC	RECEIVER CONTROLLER		
RES	RESISTOR		
RF	RETURN FAN		
RHC	REHEAT COIL		

HVAC PIPING SYMBOLS

- (HPS) HIGH PRESSURE STEAM
- (MPS) MEDIUM PRESSURE STEAM
- (LPS) LOW PRESSURE STEAM
- CR CONDENSATE RETURN
- (MU) MAKEUP WATER
- (HWS) LOW TEMP HOT WATER SUPPLY
- (HWR) LOW TEMP HOT WATER RETURN
- (MTWS) MEDIUM TEMP HOT WATER SUPPLY
- (MTWR) MEDIUM TEMP HOT WATER RETURN
- (HTWS) HIGH TEMP HOT WATER SUPPLY
- (HTWR) HIGH TEMP HOT WATER RETURN
- (RD) REFRIGERANT DISCHARGE
- (RS) REFRIGERANT SUCTION
- (RL) REFRIGERANT LIQUID
- (CWS) CONDENSER WATER SUPPLY
- (CWR) CONDENSER WATER RETURN
- (CHWS) CHILLED WATER SUPPLY
- (CHWR) CHILLED WATER RETURN
- (HCS) HOT/CHILLED WATER SUPPLY
- (HCR) HOT/CHILLED WATER RETURN
- (DHWS) DOMESTIC HOT WATER SUPPLY
- (DHWC) DOMESTIC HOT WATER RECIRCULATION

LINE TYPES

- P — PNEUMATIC TUBE
- 120VAC WIRING (FACTORY/PANEL)
- 120VAC WIRING (FIELD)
- 24VAC WIRING (FACTORY/PANEL)
- 24VAC WIRING (FIELD)
- #### LOW VOLTAGE CABLE (FIELD)
- COLOR COMMUNICATION TRUNK
  - CABLE'S PHYSICAL DESCRIPTION

CABLE SPECIFICATIONS

COMMUNICATIONS
ECHELON (I/A) – 22/2 PLENUM/NO SHIELD/ORANGE
ASD BUS (NW8) – 22/2 PLENUM/SHIELDED/BLUE
MICROSMART BUS (DMS) – 22/2 PLENUM/SHIELDED/BLUE
MSTP (KMC & TAC) – 24/2 PLENUM/SHIELDED/WHITE WITH RED STRIPE
MODBUS – 22/2 PLENUM/SHIELDED/GREEN
MICRONET U–BUS – 22/2 PLENUM/NO SHIELD/GREY

CONTROLLER INPUTS AND OUTPUTS
(CONSISTING OF ALL CONTROLLERS EXCEPT MICRONET DEVICES)
18/2 PLENUM/SHIELDED/YELLOW
18/2 PLENUM/SHIELDED/TAN
18/2 PLENUM/SHIELDED/CREAM
18/2 PLENUM/SHIELDED/LIGHT PURPLE
18/2 PLENUM/SHIELDED/LIGHT BLUE
18/2 PLENUM/SHIELDED/TEAL
18/2 PLENUM/SHIELDED/PINK
NETSENSOR (KMC) – 24/6 PLENUM/NO SHIELD/BROWN

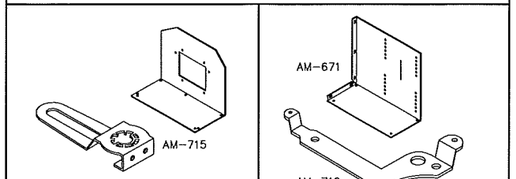
MICRONET CONTROLLER INPUTS AND OUTPUTS
U–LINK (SENSOR) – 22/2 PLENUM/NO SHIELD/PURPLE
FLOATING VALVE – 18/3 PLENUM/ NO SHIELD/WHITE
MICRONET POWER – 16/2 PLENUM/NO SHIELD/WHITE

NOTE: WIRES CARRYING AN EXTRNAL POWER SOURCE THAT ENTER OUR TCP's WILL BE YELLOW FOR ALL APPLICATIONS

DRAWING SYMBOLS

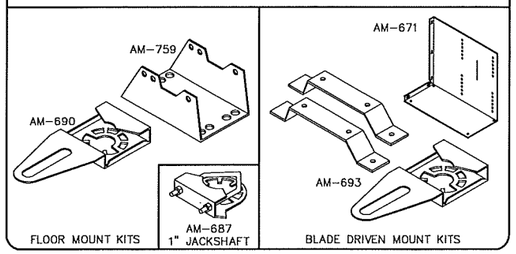
- TAG — DEVICE IDENTIFICATION NAME
- 99 — REFER TO EXPLANATION IN "NOTES"
- A — DRAWING AFFECTED BY REVISION
- A / XXX — AREA AFFECTED BY REVISION
- CONTROL PANEL TERMINAL NUMBER
- AA — REFERENCE TO DETAIL DRAWING
- # — CONTROL PANEL PNEUMATIC TERMINAL NUMBER
- M — 20 PSIG MAIN AIR SUPPLY
- 16 DAY / 20 NIGHT — DUAL PRESSURE AIR MAIN

35 IN-LB DURADRIVE HARDWARE



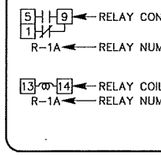
FLOOR MOUNT KITS — BLADE DRIVEN MOUNT KITS

60 AND 133 IN-LB DURADRIVE HARDWARE

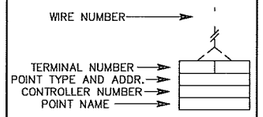


FLOOR MOUNT KITS — BLADE DRIVEN MOUNT KITS

RELAY TERMINAL DESCRIPTION



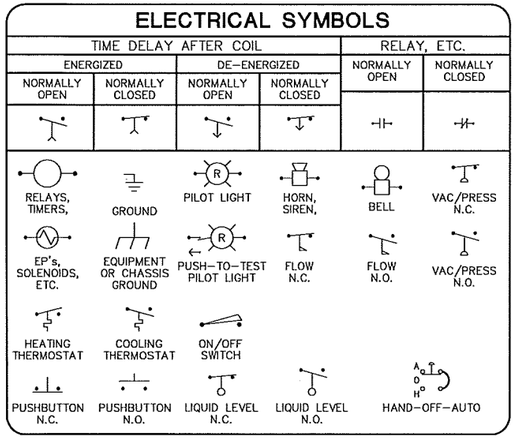
DDC POINT DESCRIPTION



PNEUMATIC TUBING SPECIFICATIONS

COLOR	USE
BLACK	20PSI MAIN AIR
WHITE	PNUEMATIC BRANCH LINE
RED	FIRE DAMPERS OR EMERGENCY MAIN AIR OR HOT WATER VALVES
ORANGE	AIR HANDLING UNIT DAMPERS
GREEN	OUTSIDE AIR STATIC PRESSURE REFERENCE
BLUE	CHILLED WATER VALVE
PURPLE	DUCT STATIC PRESSURE REFERENCE
YELLOW	BUILDING STATIC PRESSURE REFERENCE
TWIN TUBE	PLAIN TUBE IS MAIN AIR; TUBE WITH LABELING IS PNEUMATIC BRANCH LINE

ELECTRICAL SYMBOLS



INPUT AND OUTPUT RESISTORS

DMS – MICROSMART CONTROLLERS
1K, 1%, 1/4WATT RESISTOR – CONVERTS ANALOG INPUT TO DIGITAL INPUT.
500 OHM, 1%, 1/4 WATT RESISTOR – CONVERTS 4–20mA OUTPUT TO 0–10VDC.
750 OHM, 1%, 1/4 WATT RESISTOR – CONVERTS 4–20mA OUTPUT TO 0–15VDC.

R500 SERIES ONLY
22.1 OHM, 1%, 1/4 WATT RESISTOR – ON 4–20mA INPUTS.

NETWORK 8000 (MICROZONE II)
250Ω 250 OHM, 1%, 1/4 WATT RESISTOR – CONVERTS 4–20mA INPUT TO 1–5VDC.
500Ω 500 OHM, 1%, 1/4 WATT RESISTOR – CONVERTS 4–20mA OUTPUT TO 0–10VDC.

INSTALLATION NOTES:

PNEUMATIC INSTALLATION NOTES:
ALL EXPOSED PNEUMATIC TUBING WILL BE POLYETHYLENE TUBING ENCLOSED IN METAL RACEWAY, OR WILL BE SEAMLESS HARD DRAWN COPPER. ALL CONCEALED PIPING WILL BE SEAMLESS SOFT DRAWN COPPER, HARD DRAWN COPPER OR POLYETHYLENE TUBING ENCLOSED IN CONDUIT. POLYETHYLENE FIRE RATED TUBING MAY BE RUN IN CONCEALED AND FULLY ACCESSIBLE AREAS PROVIDED THERE IS NO POSSIBILITY OF PHYSICAL DAMAGE. COPPER HARD DRAWN TUBING INSTALLATIONS EXPOSED TO VIEW WILL HAVE A MAXIMUM UNSUPPORTED LENGTH OF 3 FEET, (7 FEET IN ALL OTHER LOCATIONS). A SUPPORT WILL BE PROVIDED WITHIN 12 INCHES OF EACH CHANGE OF DIRECTION. POLYETHYLENE TUBING WILL BE ROUTED IN CABLE TRAY, METALLIC RACEWAY, ATTACHED TO CONDUIT DEDICATED FOR THAT SOLE PURPOSE, OR SUPPORT WIRES THAT PROVIDE SECURE SUPPORT AND THAT ARE INSTALLED IN ADDITION TO THE CEILING GRID SUPPORT WIRES. TUBING SHALL NOT BE ATTACHED TO PIPES OR CONDUIT INSTALLED BY OTHERS OR CEILING GRID SUPPORT WIRES. TUBING WILL BE INSTALLED PARALLEL WITH BUILDING LINES. TUBING SHALL NOT BE LOOSELY DRAPED BETWEEN SUPPORTS OR LAID ACROSS CEILING TILES AND WILL NOT BE INSTALLED IN THE SAME CONDUIT AS WIRING. SMALL AMOUNTS OF SLACK MAY BE LEFT WHEN CONNECTING TO CONTROL PARTS SUCH AS ACTUATORS OR WHERE VIBRATION DAMAGE MAY OCCUR. TUBING USED IN A CONCRETE POUR MUST BE INSTALLED IN METALLIC CONDUIT OR RIDGED NONMETALLIC CONDUIT.

WIRING INSTALLATION NOTES:
INSTALLATION OF WIRING, CABLE, CONDUIT ETC. WILL CONFORM TO DIVISION 16 REQUIREMENTS. PLENUM WIRING MAY BE INSTALLED IN ACCESSIBLE AREAS WHERE NOT EXPOSED TO VIEW OR SUBJECT TO PHYSICAL DAMAGE. PLENUM WIRING WILL NOT CARRY ANY VOLTAGE GREATER THAN 24 VOLTS. PLENUM WIRE OR CABLE WILL BE ROUTED IN CABLE TRAY, METALLIC RACEWAY, ATTACHED TO CONDUIT DEDICATED FOR THAT SOLE PURPOSE, OR SUPPORT WIRES THAT PROVIDE SECURE SUPPORT AND THAT ARE INSTALLED IN ADDITION TO THE CEILING GRID SUPPORT WIRES. WIRING SHALL NOT BE ATTACHED TO PIPES OR CONDUIT INSTALLED BY OTHERS OR CEILING GRID SUPPORT WIRES. WIRING WILL BE INSTALLED PARALLEL WITH BUILDING LINES AND SECURED AT LENGTHS NO GREATER THAN 4 1/2 FEET. PLENUM WIRING SHALL MAINTAIN A MINIMUM SPACING OF 6 INCHES FROM ANY LINE VOLTAGE CONDUIT OR DEVICE. PLENUM COMMUNICATION CABLE SHALL BE ROUTED NO CLOSER THAN 12 INCHES FROM ANY FLUORESCENT OR HIGH DISCHARGE ELECTRICAL LIGHTING AND 3 FEET FROM VOLTAGES AND TRANSFORMERS GREATER THAN 600 VOLTS. PLENUM WIRING WITH OR WITHOUT SHIELD WILL NOT BE SPLICED OUTSIDE OF CONTROL DEVICES UNLESS ABSOLUTELY NECESSARY THEN IT MUST BE DONE WITHIN AN APPROVED ENCLOSURE CONTAINING TERMINAL BLOCK CONNECTIONS LOCATED IN AN ACCESSIBLE AREA. COMMUNICATION WIRING WILL NOT BE RUN IN THE SAME CONDUIT OR ENCLOSURE AS LINE VOLTAGE WIRING WITH THE EXCEPTION OF INSIDE D.D.C. CONTROL PANELS. CONTROL WIRING AND LINE VOLTAGE WIRING WILL BE ALLOWED TO OCCUPY THE SAME ENCLOSURE ONLY WHEN FUNCTIONALLY ASSOCIATED FOR THE CONTROL OF THE LINE VOLTAGE EQUIPMENT.

ALL APPLICABLE CODES AND SPECIFICATIONS SHALL APPLY. THOSE MORE STRINGENT THAN THE ABOVE GUIDELINES SHALL TAKE PRECEDENCE.

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CITY OF COLORADO SPRINGS - EPC (CITY)
DRAWING LEGEND
30 SOUTH NEVADA AVENUE
COLORADO SPRINGS, COLORADO 80901

NO.	Revision:	Date:	By:
A			
B			
C			
D			
E			
F			

Engineer: BURNS & MCDONNELL
Contractor: BURNS & MCDONNELL
Proj. Mgr.: B. STUBER
App. Eng.: B. DHAKAL
Created: 08-22-07
Standard Template: N/A

JOB NUMBER TC2955
SHEET NO. 2

VALVE SCHEDULE

TAG	QTY	SIGNAL	LOCATION	PART #	MFG	SIZE	PATTERN	FAILSAFE POSITION	GPM	#/HR	CALC.	CV	PD (PSI)
UHV-1	1	2-10VDC	UH-16	VS-7213-804-4-09	TAC	1-1/4"	2-WAY	N.O.	33.00		14.76	20.00	2.7
UHV-2	1	2-10VDC	UH-17	VS-7213-804-4-09	TAC	1-1/4"	2-WAY	N.O.	40.60		18.25	20.00	4.2
BBRV-1	1	24VAC	1ST FLOOR SOUTH	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-2	1	24VAC	1ST FLOOR E& SE	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-3	1	24VAC	1ST FLOOR WEST	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-4	1	24VAC	2ND FLOOR SOUTH	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-5	1	24VAC	2ND FLOOR E& SE	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-6	1	24VAC	2ND FLOOR WEST	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-7	1	24VAC	3RD FLOOR SOUTH	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-8	1	24VAC	3RD FLOOR E& SE	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-9	1	24VAC	3RD FLOOR WEST	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-10	1	24VAC	3RD FLOOR N& NE	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-11	1	24VAC	4TH FLOOR SOUTH	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-12	1	24VAC	4TH FLOOR E& SE	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-13	1	24VAC	4TH FLOOR WEST	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-14	1	24VAC	4TH FLOOR N& NE	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-15	1	24VAC	5TH FLOOR SOUTH	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-16	1	24VAC	5TH FLOOR E& SE	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-17	1	24VAC	5TH FLOOR WEST	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-18	1	24VAC	5TH FLOOR N& NE	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-19	1	24VAC	6TH FLOOR SOUTH	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-20	1	24VAC	6TH FLOOR E& SE	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-21	1	24VAC	6TH FLOOR WEST	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-22	1	24VAC	6TH FLOOR N& NE	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-23	1	24VAC	7TH FLOOR SOUTH	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-24	1	24VAC	7TH FLOOR E& SE	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-25	1	24VAC	7TH FLOOR WEST	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					
BBRV-26	1	24VAC	7TH FLOOR N& NE	B220+TFX24	BELIMO	3/4"	2-WAY	N.O.					

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VALVE SCHEDULE
30 SOUTH NEVADA AVENUE
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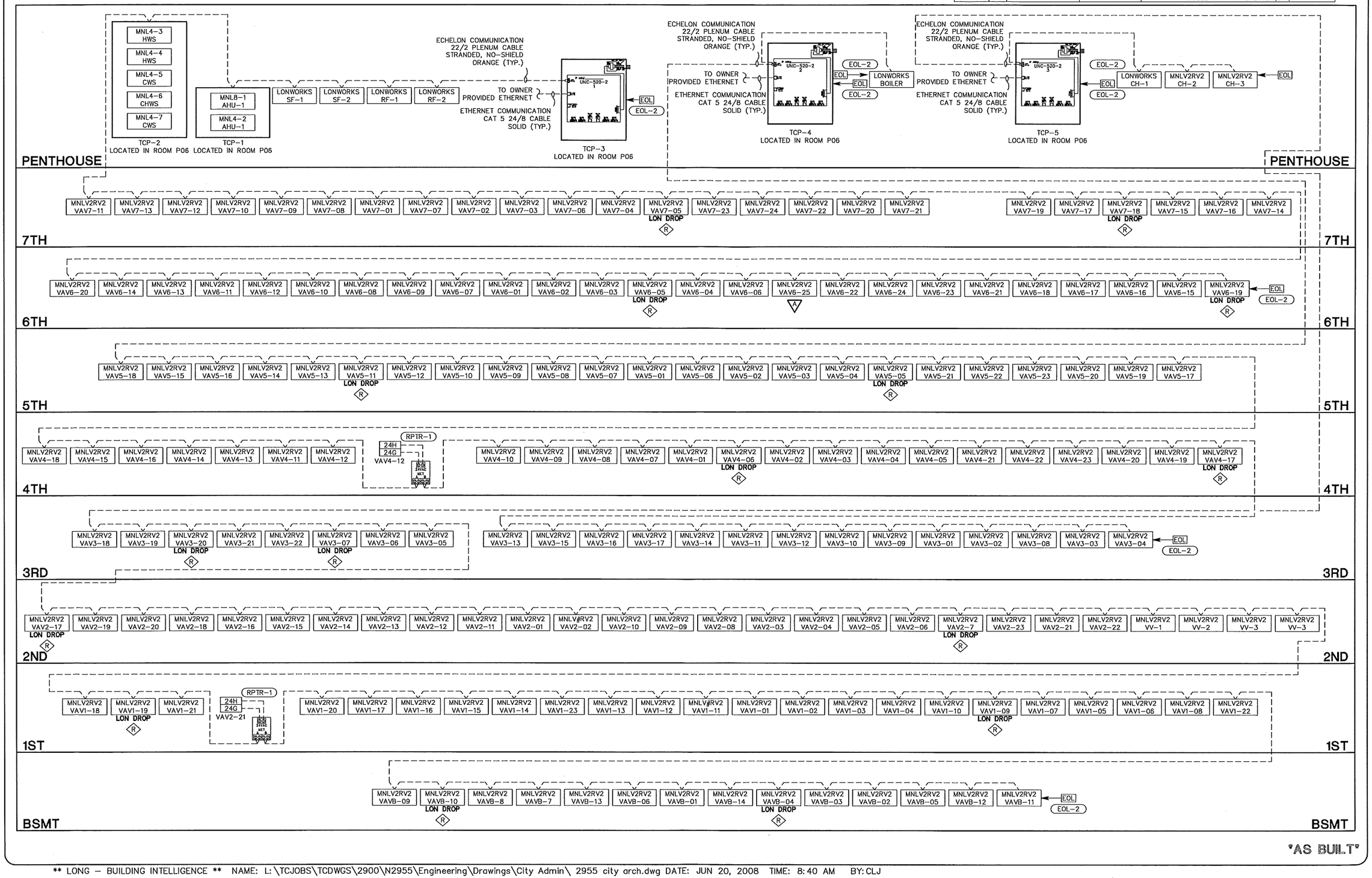
NO.	Revision:	Date:	By:
A	TC2955 AS BUILT	08-20-08	CLJ
B			
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D			
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Engineer: BURNS & MCDONNELL
Contractor:BURNS & MCDONNELL
Proj. Mgr.: B. STUBER
App. Eng.: B. DHAKAL
Created: 08-22-07
Standard Template: N/A

JOB NUMBER TC2955
SHEET NO. 3

BILL OF MATERIAL

Tag	Qty	Part #	Supplier	Description
EOL-2	6	LON-TERM2	TAC	EOL FOR DAISY CHAIN TOPOLOGY
RPTR-1	2	0110-00	NEUROLOGIC	MODEL 110 FTT-10 REPEATER (LON)



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Denver 303-375-2100
Colorado Springs 719-592-4364
Wyoming 307-265-5997

CITY OF COLORADO SPRINGS - EPC (CITY)

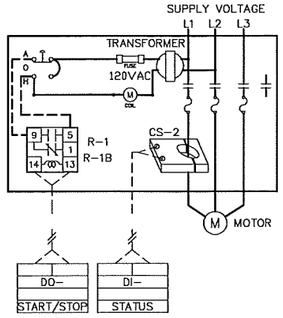
SYSTEM ARCHITECTURE
30 SOUTH NEVADA AVENUE
COLORADO SPRINGS, COLORADO 80901

NO.	Revision:	Date:	By:
A	ENGINEERING REVIEW	10-12-07	BD
B	TC3082 AS BUILT	03-28-08	JJV
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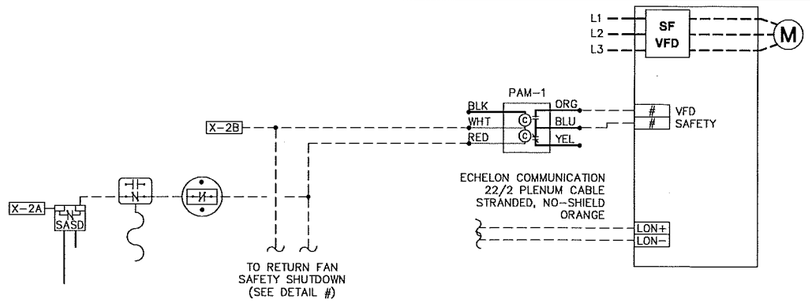
Engineer: BURNS & MCDONNELL
Contractor: BURNS & MCDONNELL
Proj Mgr.: B. STUBER
App. Eng.: B. DHAKAL
Created: 08-22-07
Standard Template: N/A

JOB NUMBER TC2955
SHEET NO. 4

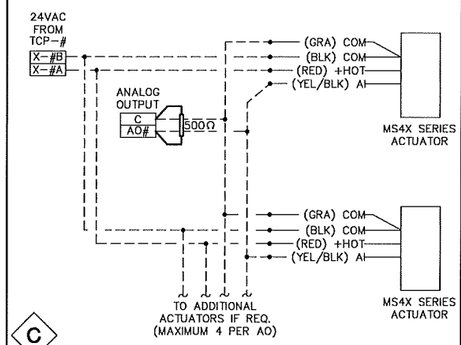
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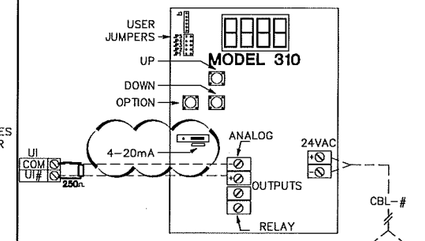
A THREE PHASE STARTER W/ DDC CONTROL



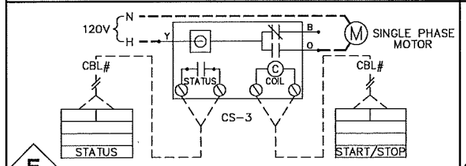
B TYPICAL AHU SF VFD W/ DDC CONTROL AND SAFETIES DETAIL



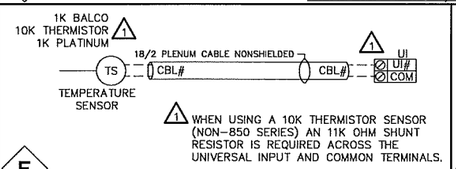
C 2-10VDC DURADRIVE ACTUATOR



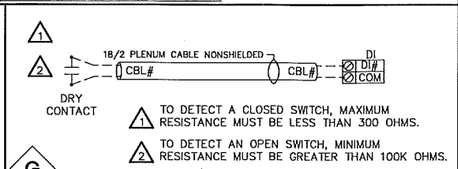
D DCS MODEL 310 CARBON DIOXIDE SENSOR DETAIL



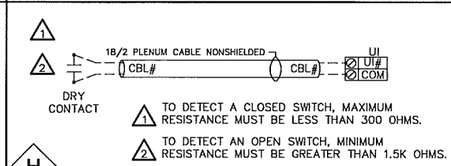
E SINGLE PHASE MOTOR WITH DIRECT DIGITAL CONTROL



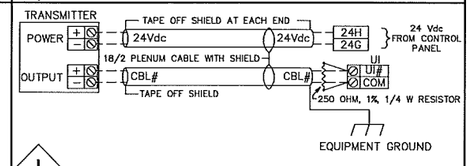
F RTD SENSOR TO "I/A" UNIVERSAL INPUT



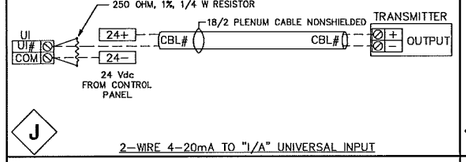
G DRY CONTACT TO "I/A" DIGITAL INPUT



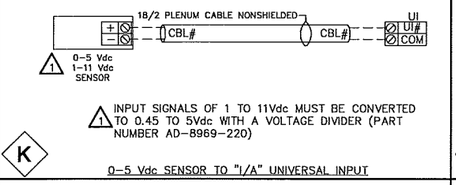
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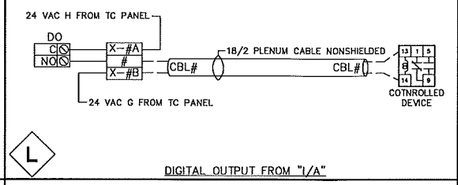
I 4-WIRE 4-20mA TO "I/A" UNIVERSAL INPUT



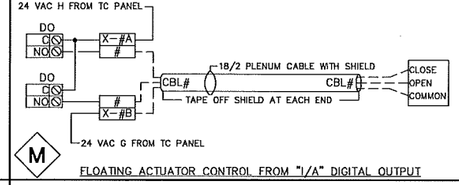
J 2-WIRE 4-20mA TO "I/A" UNIVERSAL INPUT



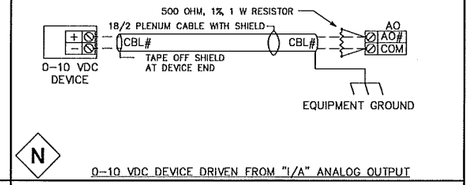
K 0-5 Vdc SENSOR TO "I/A" UNIVERSAL INPUT



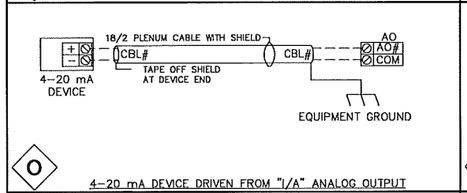
L DIGITAL OUTPUT FROM "I/A"



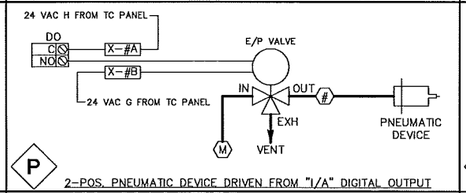
M FLOATING ACTUATOR CONTROL FROM "I/A" DIGITAL OUTPUT



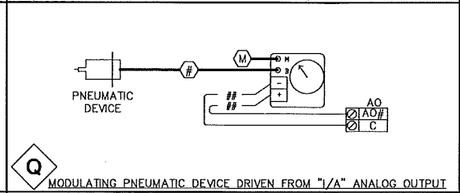
N 0-10 VDC DEVICE DRIVEN FROM "I/A" ANALOG OUTPUT



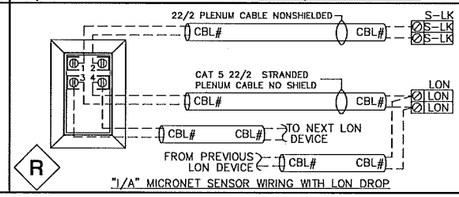
O 4-20 mA DEVICE DRIVEN FROM "I/A" ANALOG OUTPUT



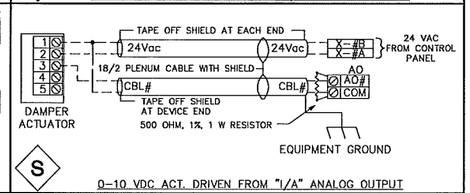
P 2-POS. PNEUMATIC DEVICE DRIVEN FROM "I/A" DIGITAL OUTPUT



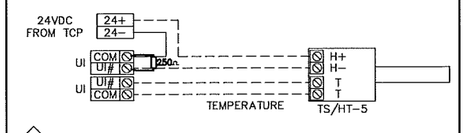
Q MODULATING PNEUMATIC DEVICE DRIVEN FROM "I/A" ANALOG OUTPUT



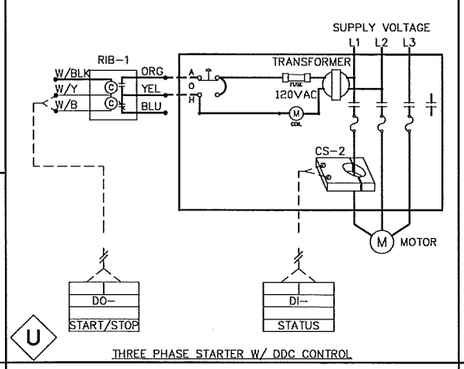
R "I/A" MICRONET SENSOR WIRING WITH LON DROP



S 0-10 VDC ACT. DRIVEN FROM "I/A" ANALOG OUTPUT



T KELE HO30K-T85 SERIES DETAIL



U THREE PHASE STARTER W/ DDC CONTROL

"AS BUILT"

LONG
Wyoming 307-265-5997
Colorado Springs 719-592-4364
Denver 303-975-2100

CITY OF COLORADO SPRINGS - EPC (CITY)
DRAWING DETAILS
30 SOUTH NEVADA AVENUE
COLORADO SPRINGS, COLORADO 80901

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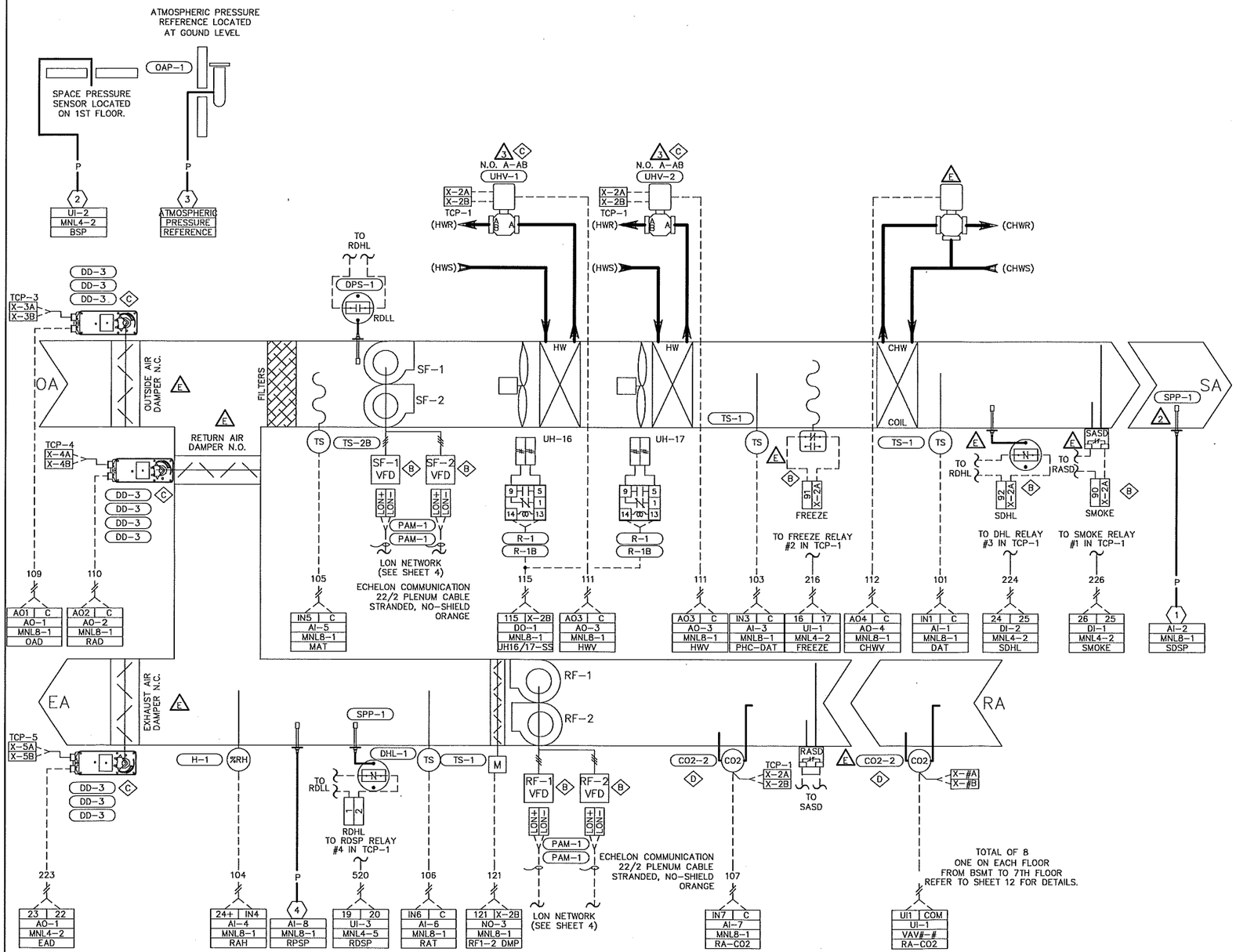
Engineer: BURNS & MCDONNELL
Contractor:BURNS & MCDONNELL
Proj Mgr.: B. STUBER
App. Eng.: B. DHAKAL
Created: 08-22-07

Standard Template: N/A

JOB NUMBER TC2955
SHEET NO. 5

NOTES:

2. FIELD VERIFY LOCATION 2/3rds WAY DOWN MAIN SUPPLY DUCT.
3. REFER TO VALVE SCHEDULE ON SHEET 3.
E. EXISTING DEVICE.



AHU-#

REFER TO SHEETS 7 AND 11 FOR PANEL TERMINATIONS.
NO SUBMITTAL RECEIVED. FIELD VERIFY WIRING.

"AS BUILT"

BILL OF MATERIAL

Tag	Qty	Part #	Supplier	Description
CO2-2	9	DCS-310ED	TAC	CARBON DIOXIDE MONITOR (DUCT MOUNT)
DA-1	8	MK-7121	TAC	8-13# DAMPER ACTUATOR
DHL-1	1	1900-5-MR	DWYER	DUCT HIGH PRESSURE LIMIT (MANUAL RESET)
DPS-1	1	1910-5	DWYER	DIFF. PRESSURE SWITCH (FILTER)
H-1	1	HU-224-2-MA	MAMAC	HUMIDITY TRANSMITTER (DUCT)
OAP-1	1	A-306	DWYER	OUTDOOR PRESSURE REFERENCE STATION
PAM-1	4	P-PAM-1	TAC	SPDT 120/24V RELAY (10AMP)
R-1	2	RH1BULAC24	IDEC	SPDT 24VAC RELAY W/ LED
R-1B	2	SH1B05	IDEC	SPDT RELAY BASE
SPP-1	2	A-520-1-A-1	MAMAC	DUCT PRESSURE PICKUP TUBE
TS-1	3	TS-9201	TAC	DUCT SENSOR
TS-2B	1	TS-9422	TAC	AVERAGING DUCT SENSOR (22')

TC3179 BILL OF MATERIAL

Tag	Qty	Part #	Supplier	Description
DD-3	10	MS41-7153	TAC	24VAC/DC, MOD., S.R. DAMPER ACTUATOR 133 IN-LB

SEQUENCE OF OPERATION

AIR HANDLING UNIT (AHU-1):

THE TAC MNL-800 AND MNL-200 I/A SERIES DIRECT DIGITAL CONTROLLERS (DDC) PROVIDE CONTROL OF THE UNIT AS FOLLOWS.

A USER ADJUSTABLE OCCUPIED/UNOCCUPIED TIME SCHEDULE (6:00 A.M. TO 6:00 P.M.) CONTROLS THE UNIT MODE OF OPERATION.

DAMPER NORMAL POSITION:
OUTSIDE AIR DAMPER NORMALLY CLOSED.
RETURN AIR DAMPER NORMALLY OPEN.
EXHAUST AIR DAMPER NORMALLY CLOSED.

AIR HANDLING UNIT OFF:
THE SUPPLY FANS AND RETURN FANS ARE OFF, THE OUTSIDE AIR DAMPER AND EXHAUST DAMPER ARE CLOSED, THE RETURN AIR DAMPER IS OPEN, AND THE COOLING COIL CONTROL VALVE IS CLOSED. THE HEATING COIL CONTROL VALVES ARE OPEN.

OCCUPIED MODE
THE DDC ENABLES THE SUPPLY FANS AND RETURN FANS TO RUN CONTINUOUSLY. DURING THE INITIALIZATION PERIOD OF 5 MINUTES (ADJ.) THE DDC DELAYS THE OPENING OF OUTSIDE AIR DAMPERS TO ALLOW A STABILIZATION PERIOD FOR THE AHU. DURING THIS PERIOD THE HEATING CONTROL VALVES WILL BE UNDER DISCHARGE AIR TEMPERATURE CONTROL, THE CHILLED WATER VALVE AND OUTSIDE AIR DAMPER WILL BE CLOSED AND THE RETURN AIR DAMPER WILL BE OPEN. THE VAV BOXES WILL BE POSITIONED TO MAXIMUM FLOW. THE DDC MODULATES THE SUPPLY FAN VARIABLE FREQUENCY DRIVES (VFD) TO MAINTAIN THE SUPPLY DUCT STATIC PRESSURE SET POINT OF 1.0" W.C. (ADJ.) AS MEASURED BY DIFFERENTIAL PRESSURE TRANSMITTER (DPT-2). THE DDC MODULATES THE RETURN FAN VFDS TO MAINTAIN THE BUILDING STATIC PRESSURE (DPT-1) SET POINT OF 0.05" WC (ADJ.). THE DDC MONITORS THE STATUS OF THE FANS. IF FANS FAIL TO PROVIDE RUN STATUS WHILE COMMANDED "ON" (S/S=ON AND STATUS=OFF) THE DDC RETURNS THE DAMPERS TO THEIR NORMAL POSITIONS AND AN ALARM IS GENERATED AT THE DDC CENTRAL WORKSTATION. AT THE COMPLETION OF THE INITIALIZATION PERIOD THE VAV BOXES ARE RELEASED TO AUTOMATIC CONTROL ONE FLOOR AT A TIME WITH A 2 MINUTE (ADJ.) DELAY BETWEEN FLOORS, THE AHU DAMPERS, CHILLED WATER VALVE, AND HEATING VALVES ARE RELEASED TO CONTROL AS DESCRIBED BELOW.

THE DDC MONITORS THE RETURN DUCT STATIC PRESSURE (DPT-1). DURING THE ECONOMIZER MODE THE BUILDING PRESSURE CONTROL IS OVERRIDDEN WHEN THE LOW RETURN AIR PRESSURE (0.0 " WC ADJ.) IS MEASURED. THE RETURN FAN VFDS RAMP UP TO MAINTAIN A SET POINT OF 0.15" WC (ADJ.).

THE DDC MODULATES THE RETURN AIR AND OUTSIDE AIR DAMPERS INVERSELY TO PROVIDE AIRSIDE ECONOMIZER OPERATION WHENEVER THERE IS A CALL FOR COOLING AND THE OUTSIDE AIR ENTHALPY IS LOWER THAN THE RETURN AIR ENTHALPY. DURING THIS TIME, THE DDC MODULATES THE DAMPERS INVERSELY TO MAINTAIN THE MIXED AIR TEMPERATURE (TS-2B) EQUAL TO THE DISCHARGE AIR TEMPERATURE SET POINT AND MODULATES THE EXHAUST AIR DAMPER IN PARALLEL WITH THE OUTSIDE AIR DAMPERS, MINUS 20% (ADJ.). THE ECONOMIZER OPERATION IS DISABLED WHEN OUTSIDE AIR ENTHALPY IS GREATER THAN RETURN AIR ENTHALPY OR WHENEVER THE OUTSIDE AIR TEMPERATURE (TS/HT-5) IS BELOW 40°F (ADJ.).

WHEN THE OUTSIDE AIR TEMPERATURE IS BELOW 40°F (ADJ.), THE EXHAUST AIR DAMPER REVERTS TO CONTROL VIA BUILDING DIFFERENTIAL PRESSURE, CLOSE AT 0.05" WC, OPEN AT 0.1" WC.

THE DDC RESETS THE DISCHARGE AIR TEMPERATURE FROM 55°F TO A HIGHER SET POINT OF 60°F (ADJ.) IF 2 OR MORE (ADJ.) INTERIOR ZONES CALL FOR HEAT AS INDICATED BY A MINIMUM VAV BOX AIR FLOW AND SPACE TEMPERATURE (S3) BELOW THE SET POINT. IF AFTER 15 MINUTES (ADJ.) THE INTERIOR ZONES ARE STILL BELOW SET POINT, THE DDC RESETS THE DISCHARGE AIR TEMPERATURE TO 65°F. AS INTERIOR ZONES ARE SATISFIED, THE DISCHARGE AIR TEMPERATURE RESET SEQUENCE IS REVERSED. IF THE COOLING COIL ENTERING TEMPERATURE (TS-1) IS BELOW THE DISCHARGE AIR TEMPERATURE SET POINT, THE DDC MODULATES THE PRE-HEAT CONTROL VALVES (UHV-1 &2) TO MAINTAIN SET POINT.

THE DDC MODULATES THE OUTSIDE AIR AND RETURN AIR DAMPERS INVERSELY TO MAINTAIN THE REQUIRED MINIMUM FLOW OF OUTSIDE VENTILATION AIR TO MAINTAIN THE CO2 LEVEL AT ALL SENSOR LOCATIONS TO NO MORE THAN 800 PPM (ADJ.) MINIMUM VENTILATION IS DETERMINED BY MONITORING CO2 LEVELS IN THE OCCUPIED SPACES AT THE INLET OF EACH FLOORS RESPECTIVE RETURN AIR SHAFT AND THE COMMON RETURN AT THE UNIT.

UNOCCUPIED MODE:
THE SUPPLY FANS AND RETURN FANS ARE OFF, THE OUTSIDE AIR DAMPER AND EXHAUST DAMPER ARE CLOSED, THE RETURN AIR DAMPER IS OPEN, AND THE COOLING COIL CONTROL VALVE IS CLOSED. THE HEATING COIL CONTROL VALVES ARE OPEN. UNOCCUPIED HEATING IS ACCOMPLISHED VIA THE ZONE BASEBOARD AND UNIT HEATERS. SHOULD THE AVERAGE OF ANY ZONE FALL BELOW 68°F (ADJ.) THE BASEBOARD ZONE VALVE IS OPENED UNTIL THE AVERAGE RETURNS ABOVE 68°F

MORNING WARM UP:
AN OPTIMAL START PROGRAM IS UTILIZED TO ENABLE THE UNIT AT THE LATEST POSSIBLE TIME TO INSURE THE SPACE IS WITHIN OCCUPIED HEATING SET POINTS AT TIME OF OCCUPANCY. THE HEATING COIL CONTROL VALVES REMAIN OPEN OUTSIDE AIR AND EXHAUST AIR DAMPERS REMAIN CLOSED, AND THE RETURN AIR DAMPERS REMAIN OPEN. THESE CONDITIONS REMAIN UNTIL THE RETURN AIR TEMPERATURE (TS-1) REACHES 70°F (ADJ.). AT THAT TIME THE SYSTEM RETURNS TO NORMAL OCCUPIED OPERATION. THE SUPPLY FAN AND RETURN FAN VFDS MODULATE AS DESCRIBED IN THE OCCUPIED MODE.

UNIT SAFETIES:
A HARD WIRED LOW LIMIT THERMOSTAT (EXISTING) MOUNTED DOWNSTREAM OF THE HEATING COIL, UPON SENSING A TEMPERATURE LESS THAN 38°F (ADJ.), STOPS THE SUPPLY FANS AND RETURN FANS, CLOSES THE OUTSIDE AIR DAMPER, OPENS THE RETURN AIR DAMPERS OPENS THE HEATING COIL CONTROL VALVES, AND AN ALARM IS GENERATED AT THE DDC CENTRAL WORKSTATION.

A HARD WIRED SUPPLY DUCT HIGH LIMIT SWITCH (EXISTING) MOUNTED DOWNSTREAM OF THE SUPPLY FAN DISCHARGE, UPON SENSING A PRESSURE GREATER THAN 2" W.C. (ADJ.), STOPS THE SUPPLY FANS AND RETURN FANS, CLOSES THE OUTSIDE AIR DAMPER, OPENS THE RETURN AIR DAMPERS OPENS THE HEATING COIL CONTROL VALVES AND AN ALARM IS GENERATED AT THE DDC CENTRAL WORKSTATION.

A HARD WIRED MIXING BOX LOW LIMIT SWITCH (DPS-1) MOUNTED UPSTREAM OF THE SUPPLY FAN SUCTION, UPON SENSING A PRESSURE LESS THAN ##" W.C. (ADJ.), STOPS THE SUPPLY FANS AND RETURN FANS, CLOSES THE OUTSIDE AIR DAMPER, OPENS THE RETURN AIR DAMPERS OPENS THE HEATING COIL CONTROL VALVES AND AN ALARM IS GENERATED AT THE DDC CENTRAL WORKSTATION.

A HARD WIRED RETURN PLENUM HIGH LIMIT SWITCH (DPS-1) MOUNTED UPSTREAM OF THE RETURN FAN DISCHARGE, UPON SENSING A PRESSURE GREATER THAN #" W.C. (ADJ.), STOPS THE SUPPLY FANS AND RETURN FANS, CLOSES THE OUTSIDE AIR DAMPER, OPENS THE RETURN AIR DAMPERS OPENS THE HEATING COIL CONTROL VALVES AND AN ALARM IS GENERATED AT THE DDC CENTRAL WORKSTATION.

HARD WIRED SUPPLY AND RETURN DUCT SMOKE DETECTORS (EXISTING), UPON SENSING PRODUCTS OF COMBUSTION, STOPS THE SUPPLY FANS AND RETURN FANS, CLOSES THE OUTSIDE AIR DAMPER, OPENS THE RETURN AIR DAMPERS OPENS THE HEATING COIL CONTROL VALVES AND AN ALARM IS GENERATED AT THE DDC CENTRAL WORKSTATION

LONG
Denver 303-975-2100 | Colorado Springs 719-592-0364 | Wyoming 307-265-5997

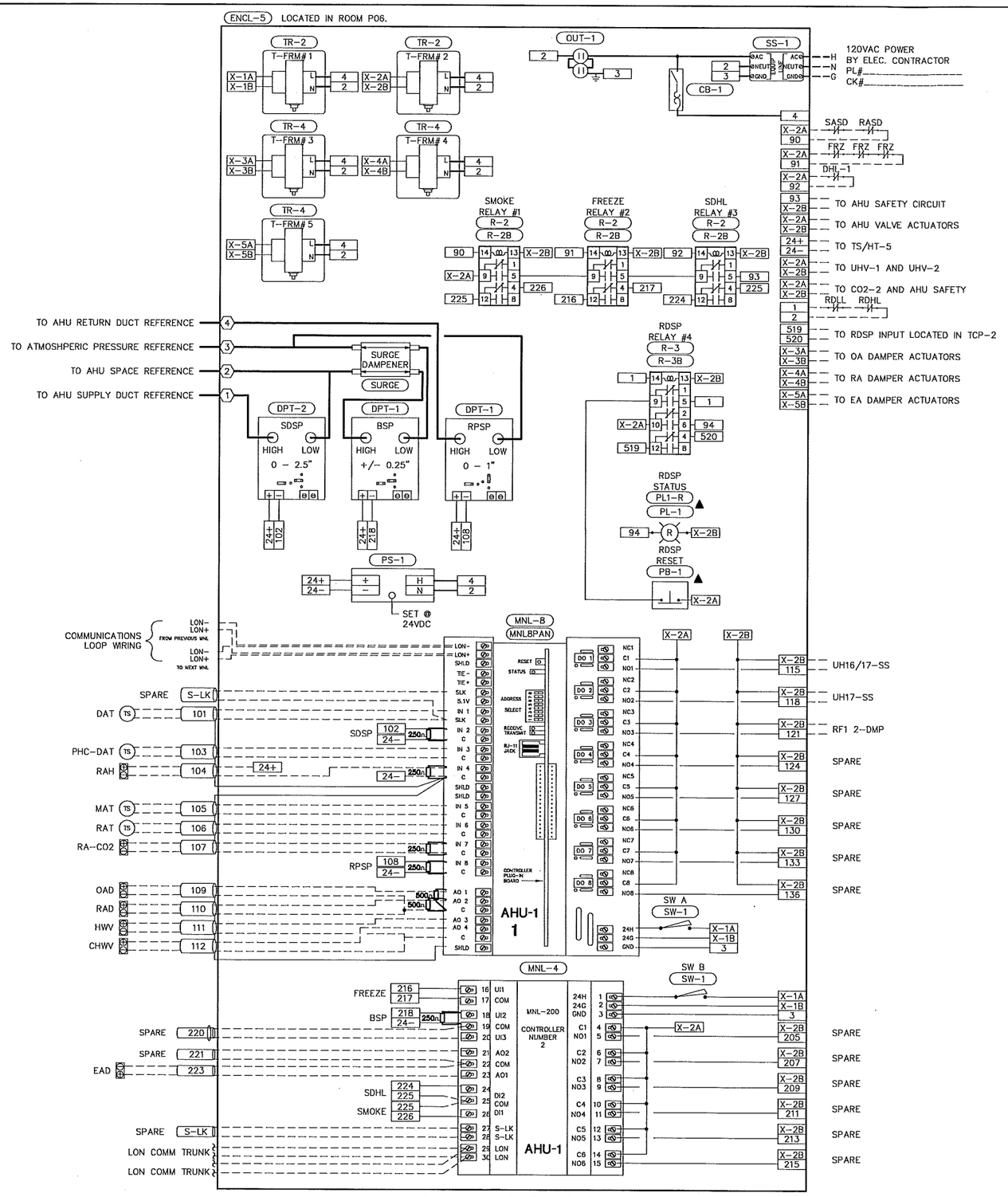
CITY OF COLORADO SPRINGS - EPC (CITY)
AHU CONTROL DIAGRAM
30 SOUTH NEVADA AVENUE
COLORADO SPRINGS, COLORADO 80901

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B	TC3179 AS BUILT	09-08-08	CSD
C	TC3179 AS BUILT	09-08-08	CSD
D	AS BUILT	11-07-08	JJV
E	AS BUILT CORRECTIONS	11-19-08	JJV
F			

Engineer: BURNS & MCDONNELL
Contractor:BURNS & MCDONNELL
Proj Mgr:: KERRY NOBLITT
App. Eng:: B. DHAKAL
Created: 08-22-07

Standard Template: TAC_LON_AHU_VFD_HC2W_ECONO_CC3W_BSP_RCO2_0_0.dwt

JOB NUMBER TC2955
SHEET NO. 6



TCP-1 AHU-1 CONTROL PANEL

"AS BUILT"

BILL OF MATERIAL

Tag	Qty	Part #	Supplier	Description
AF-1	1	K-335	TAC	PANEL AIR FILTER
CB-1	1	QZ-1-13-D-2-10	CBI	10A CIRCUIT BREAKER
DPT-1	2	PR-275-R2-MA	MAMAC	DIFF PRESS TRANSMITTER (BLDG)
DPT-2	1	PR-275-R3-MA	MAMAC	DIFF PRESS TRANSMITTER (DUCT)
ENCL-5	1	MEDIUM	PRECISION IND.	35" X 30" X 9" ENCLOSURE
EPT-1	4	CP-8551	TAC	CURRENT/PNEU TRANSDUCER
G-1	5	A201-02	TAC	0-30# GAUGE
MNL-4	1	MNL-20RS3	TAC	I/A MNL-200 CONTROLLER
MNL-8	1	MNL-800-101	TAC	I/A MNL-800 CONTROLLER CARD
MNL8PAN	1	ENCL-MZ800-PAN	TAC	I/A MNL-800 PANEL BACKPLATE
OUT-1	1	270W	COOPER	120VAC RECEPTACLE
PB-1	1	AB6M-M-1P	KELE	MOMENTARY PUSH BUTTON (SPDT)
PL-1	1	L4017	KELE	PANEL LAMP SOCKET
PL1-R	1	L4018	KELE	RED LENS
PS-1	1	PS-200-1-A-1-N	MAMAC	1.5A 24VDC POWER SUPPLY
R-2	3	RH2BULAC24	IDEC	DPDT 24VAC RELAY W/ LED
R-2B	3	SH2B05	IDEC	DPDT RELAY BASE
R-3	1	RH3BULAC24	IDEC	3PDT 24VAC RELAY W/ LED
R-3B	1	SH3B05	IDEC	3PDT RELAY BASE
SS-1	1	HSP-121-BT1RU	KELE	SURGE SUPPRESSOR
SURGE	1	SD-01	KELE	SURGE DAMPENER
SW-1	2	23F341	KELE	ON/OFF SWITCH
TR-2	2	T-208	TAC	96VA 120/24VAC TRANSFORMER

TC3179 BILL OF MATERIAL

Tag	Qty	Part #	Supplier	Description
TR-4	3	694-M2A	KELE	100VA MULTI-TAP/24VAC TRANSFORMER

NOTES:

▲ DEVICE MOUNTED ON PANEL FACE.

LONG
Wyoming 307-265-5397
Denver 303-975-2100 Colorado Springs 719-592-0364

CITY OF COLORADO SPRINGS - EPC (CITY)

TCP-1 PANEL LAYOUT
30 SOUTH NEVADA AVENUE
COLORADO SPRINGS, COLORADO 80901

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Engineer: BURNS & MCDONNELL
Contractor:BURNS & MCDONNELL
Proj Mgr.: KERRY NOBLITT
App. Eng.: B. DHAKAL
Created: 08-22-07

Standard Template: N/A

JOB NUMBER TC2955
SHEET NO. 7

NOTES:

(E) EXISTING DEVICE.

BILL OF MATERIAL

Tag	Qty	Part #	Supplier	Description
CS-2	6	CT-815	MAMAC	ADJ. SPLIT CORE CURRENT SWITCH
R-1	3	RH1BULAC24	IDEC	SPDT 24VAC RELAY W/ LED
R-1B	3	SH1B05	IDEC	SPDT RELAY BASE
RIB-1	4	RIB2401B	TAC	SPDT 120/24V RELAY IN A BOX (20AMP)
TS/HT-5	1	HO30K-785	KELE	OA HUMIDITY TRANSMITTER AND TEMP SENSOR
TS-4	6	TS-9201-106	TAC	IMMERSION SENSOR
TS4-WELL	6	SS-1	JIT MACHINE	IMMERSION SENSOR WELL

SEQUENCE OF OPERATION

HEATING WATER SYSTEM:

THE TAC MNL-800 I/A SERIES DIRECT DIGITAL CONTROLLER (DDC) AND BOILER MANUFACTURER PROVIDED LON INTERFACE PROVIDES CONTROL OF THE HEATING HOT WATER SYSTEM AS FOLLOWS.

THE PRIMARY HEATING WATER PUMPS (P-1 & 2) OPERATE ON A LEAD/STANDBY BASIS WITH ONLY ONE PUMP OPERATING AT A TIME AND THE OTHER ACTING AS A BACK UP. THE DDC AUTOMATICALLY ALTERNATES THE LEAD PUMP ON A WEEKLY BASIS TO EQUALIZE RUN TIMES.

THE DDC ENABLES THE LEAD PRIMARY HEATING WATER PUMP TO RUN CONTINUOUSLY WHEN THE OUTSIDE AIR TEMPERATURE (TS/HT-5) FALLS BELOW 65°F (ADJ.). THE DDC MONITORS THE STATUS OF THE LEAD PUMP VIA A CURRENT SWITCH (CS-2). IF THE PUMP FAILS TO PROVIDE RUN STATUS WHILE COMMANDED "ON", THE DDC ENABLES THE STANDBY PUMP AND AN ALARM IS GENERATED AT THE DDC CENTRAL WORKSTATION.

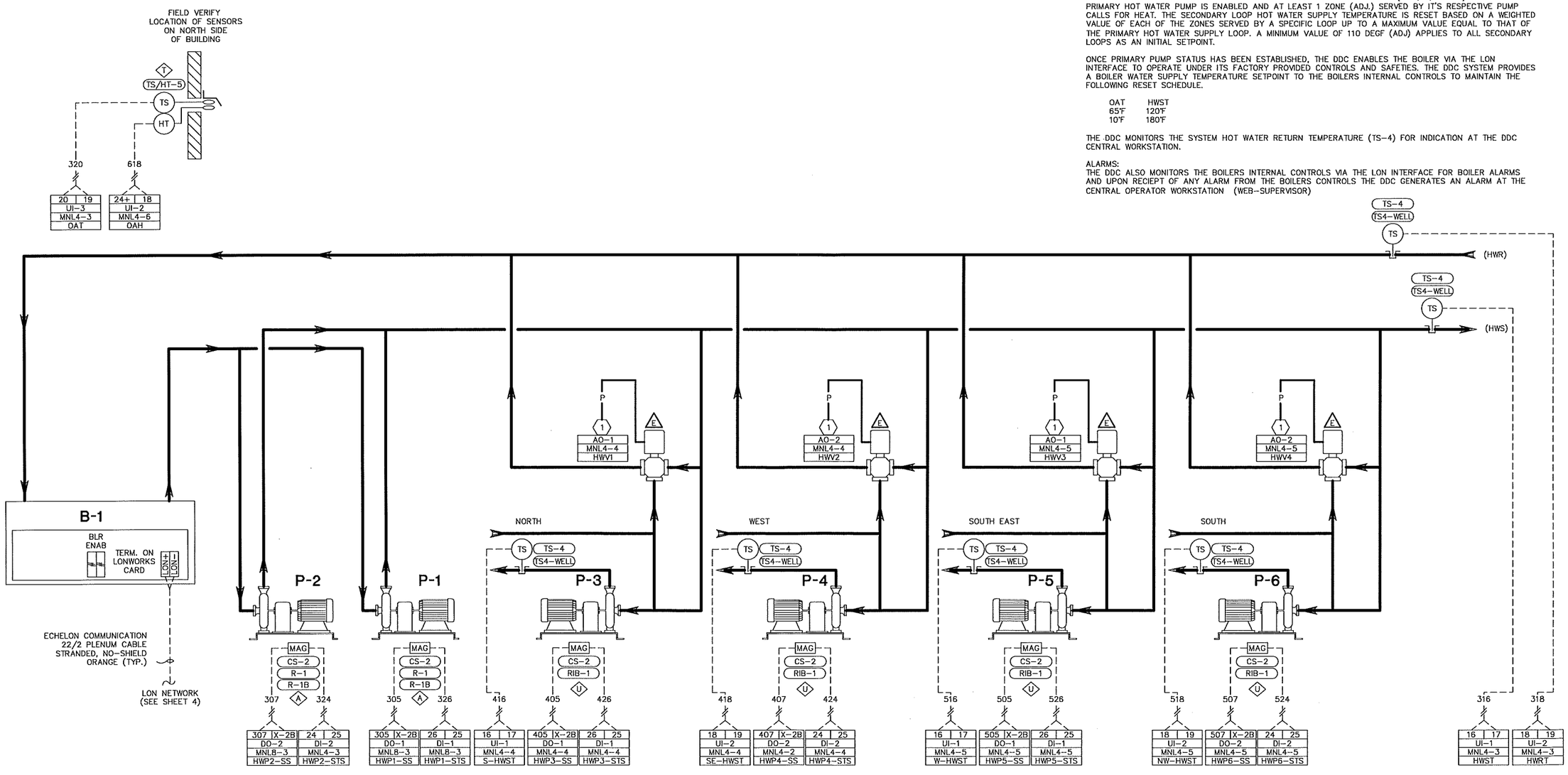
THE DDC ENABLES THE SECONDARY HOT WATER PUMP (P-3) WHENEVER THE LEAD PRIMARY HOT WATER PUMP IS ENABLED. THE DDC ENABLES THE SECONDARY HOT WATER PUMPS (P-4, 5, AND 6) WHENEVER THE PRIMARY HOT WATER PUMP IS ENABLED AND AT LEAST 1 ZONE (ADJ.) SERVED BY IT'S RESPECTIVE PUMP CALLS FOR HEAT. THE SECONDARY LOOP HOT WATER SUPPLY TEMPERATURE IS RESET BASED ON A WEIGHTED VALUE OF EACH OF THE ZONES SERVED BY A SPECIFIC LOOP UP TO A MAXIMUM VALUE EQUAL TO THAT OF THE PRIMARY HOT WATER SUPPLY LOOP. A MINIMUM VALUE OF 110 DEGF (ADJ) APPLIES TO ALL SECONDARY LOOPS AS AN INITIAL SETPOINT.

ONCE PRIMARY PUMP STATUS HAS BEEN ESTABLISHED, THE DDC ENABLES THE BOILER VIA THE LON INTERFACE TO OPERATE UNDER ITS FACTORY PROVIDED CONTROLS AND SAFETIES. THE DDC SYSTEM PROVIDES A BOILER WATER SUPPLY TEMPERATURE SETPOINT TO THE BOILERS INTERNAL CONTROLS TO MAINTAIN THE FOLLOWING RESET SCHEDULE.

OAT	HWST
65°F	120°F
10°F	180°F

THE DDC MONITORS THE SYSTEM HOT WATER RETURN TEMPERATURE (TS-4) FOR INDICATION AT THE DDC CENTRAL WORKSTATION.

ALARMS:
THE DDC ALSO MONITORS THE BOILERS INTERNAL CONTROLS VIA THE LON INTERFACE FOR BOILER ALARMS AND UPON RECIEPT OF ANY ALARM FROM THE BOILERS CONTROLS THE DDC GENERATES AN ALARM AT THE CENTRAL OPERATOR WORKSTATION (WEB-SUPERVISOR)



HEATING WATER SYSTEM

LOCATED AT ROOM P02
REFER TO SHEETS 7 AND 9 FOR PANEL TERMINATIONS.

"AS BUILT"

LONG
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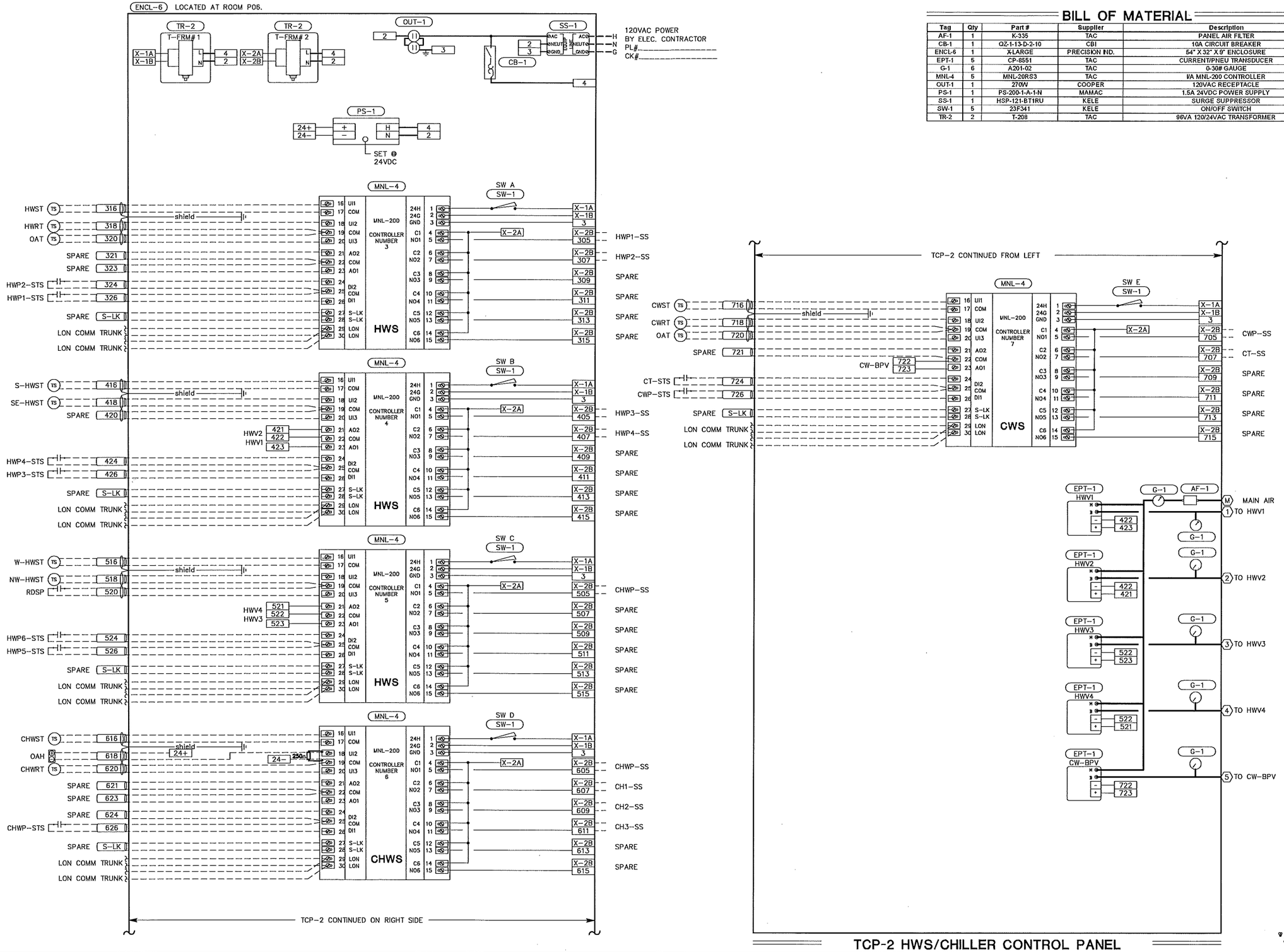
CITY OF COLORADO SPRINGS - EPC (CITY)
HEATING WATER SYSTEM CONTROL DIAGRAM
30 SOUTH NEVADA AVENUE
COLORADO SPRINGS, COLORADO 80901

NO.	Revision:	Date:	By:
A	ENGINEERING REVIEW	10-12-07	BD
B	TC2955 AS BUILT	06-20-08	CLJ
C			
D			
E			

Engineer: BURNS & MCDONNELL
Contractor: BURNS & MCDONNELL
Proj Mgr.: B. STUBER
App. Eng.: B. DHAKAL
Created: 08-22-07

Standard
Template: N/A

JOB NUMBER TC2955
SHEET NO. 8



BILL OF MATERIAL

Tag	Qty	Part #	Supplier	Description
AF-1	1	K-335	TAC	PANEL AIR FILTER
CB-1	1	QZ-1-13-D-2-10	CBI	10A CIRCUIT BREAKER
ENCL-6	1	X-LARGE	PRECISION IND.	54" X 32" X 9" ENCLOSURE
EPT-1	5	CP-8551	TAC	CURRENT/PNEU TRANSDUCER
G-1	6	A201-02	TAC	0-30# GAUGE
MNL-4	5	MNL-20RS3	TAC	I/A MNL-200 CONTROLLER
OUT-1	1	270W	COOPER	120VAC RECEPTACLE
PS-1	1	PS-200-1-A-1-N	MAMAC	1.5A 24VDC POWER SUPPLY
SS-1	1	HSP-121-BT1RU	KELE	SURGE SUPPRESSOR
SW-1	5	23F341	KELE	ON/OFF SWITCH
TR-2	2	T-208	TAC	96VA 120/24VAC TRANSFORMER

TCP-2 HWS/CHILLER CONTROL PANEL

"AS BUILT"

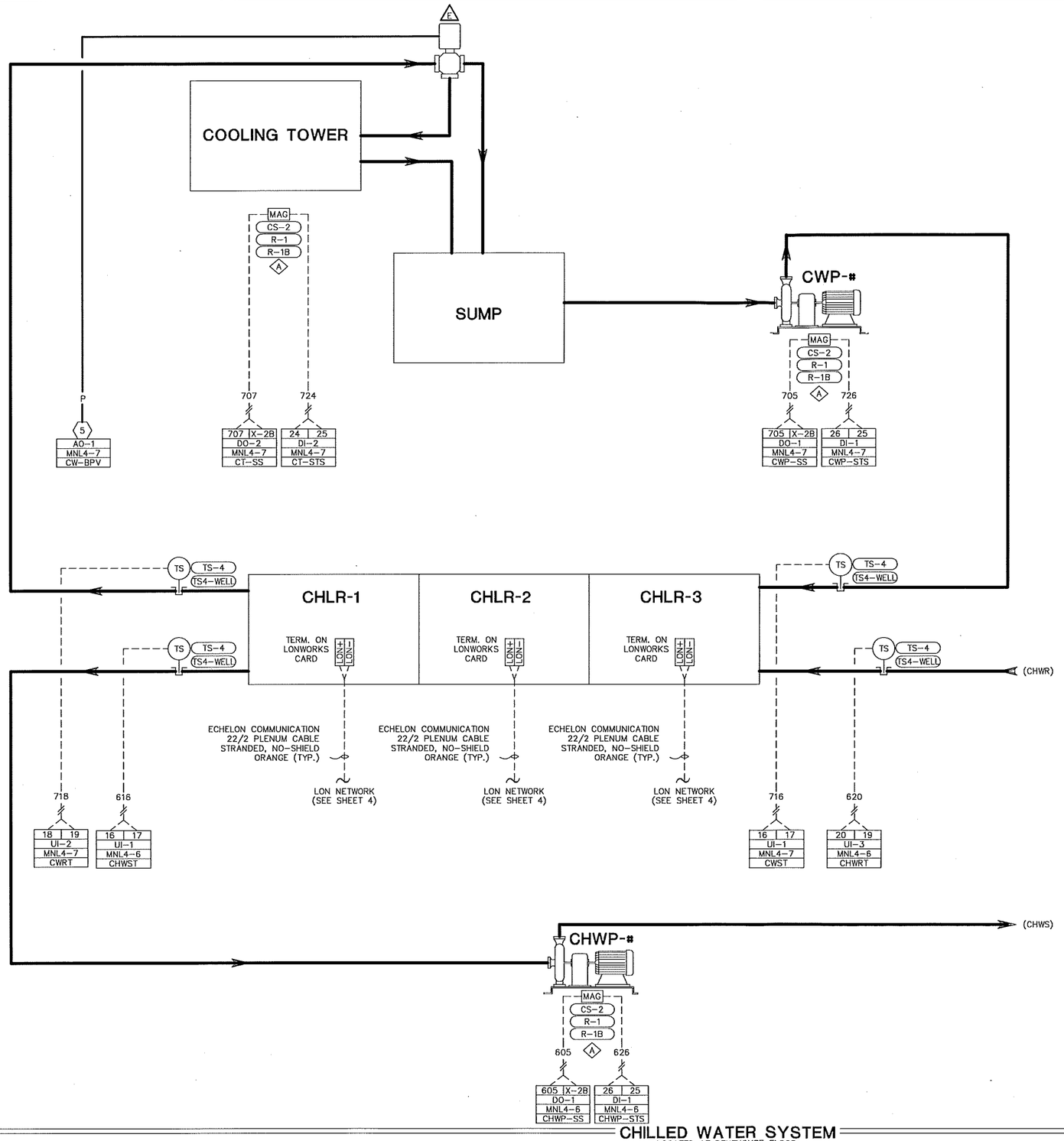
LONG
Denver 303-375-2100 Colorado Springs 719-592-4364 Wyoming 307-265-5997

CITY OF COLORADO SPRINGS - EPC (CITY)
TCP-2 PANEL LAYOUT
30 SOUTH NEVADA AVENUE
COLORADO SPRINGS, COLORADO 80901

NO.	Revision:	Date:	By:
A	TC2955 AS BUILT	06-20-08	CLJ

Engineer: BURNS & MCDONNELL
Contractor: BURNS & MCDONNELL
Proj. Mgr.: B. STUBER
App. Eng.: B. DHAKAL
Created: 08-22-07
Standard Template: N/A

JOB NUMBER TC2955
SHEET NO. 9



BILL OF MATERIAL

Tag	Qty	Part #	Supplier	Description
CS-2	3	CT-815	MAMAC	ADJ. SPLIT CORE CURRENT SWITCH
R-1	6	RH1BULAC24	IDEC	SPDT 24VAC RELAY W/ LED
R-1B	6	SH1B05	IDEC	SPDT RELAY BASE
TS-4	4	TS-9201-106	TAC	IMMERSION SENSOR
TS4-WELL	4	SS-1	JIT MACHINE	IMMERSION SENSOR WELL

SEQUENCE OF OPERATION

CHILLED WATER SYSTEM:

THE TAC MNL-200 I/A SERIES DIRECT DIGITAL CONTROLLERS (DDC) PROVIDE CONTROL OF THE SYSTEM AS FOLLOWS.

THE DDC ENABLES THE CHILLED WATER PUMP (CHWP-#) AND CONDENSER WATER PUMP (CWP-#) TO RUN CONTINUOUSLY WHEN THE OUTSIDE AIR TEMPERATURE (TS/HT-5) RISES ABOVE 55°F (ADJ.) AND THE BUILDING IS IN THE OCCUPIED MODE. THE DDC MONITORS THE STATUS OF THE PUMP VIA A CURRENT SWITCH (CS-2). IF THE PUMP FAILS TO PROVIDE RUN STATUS WHILE COMMANDED "ON", AN ALARM IS GENERATED AT THE DDC CENTRAL WORKSTATION.

ONCE THE PUMPS STATUS HAS BEEN ESTABLISHED THE DDC ENABLES THE CHILLER. ONCE ENABLED AND FLOW IS PROVEN VIA A FACTORY SUPPLIED FLOW SWITCH THE CHILLER OPERATES UNDER ITS FACTORY CONTROLS AND SAFETIES TO MAINTAIN A LEAVING WATER TEMPERATURE OF 43°F (ADJ.).

AS THE CONDENSER WATER SUPPLY TEMPERATURE (TS-4) RISES TO A VALUE EQUAL TO 5 DEGF (ADJ.) LESS THAN CONDENSOR WATER SUPPLY SETPOINT THE COOLING TOWER BYPASS VALVE MODULATES TO FULL FLOW OVER THE COOLING TOWER. AS THE CONDENSOR WATER SUPPLY TEMPERATURE CONTINUES TO RISE TO SETPOINT THE COOLING TOWER FAN IS CYCLED TO MAINTAIN CONDESOR WATER SUPPLY TEMPERATURE (+/- 3 DEGF ADJ.) AT SETPOINT FOLLOWING SCHEDULE BELOW. (CHECK ENTHALPY REQUIREMENT)

OAT	CWST
65°F	70°F
90°F	85°F

THE DDC MONITORS THE CHILLED WATER AND THE CONDENSER WATER RETURN TEMPERATURES (TS-4) FOR INDICATION AT THE DDC CENTRAL WORKSTATION.

ALARMS:
SHOULD THE CHILLED WATER PUMP, CONDENSOR WATER PUMP, OR COOLING TOWER FAN STATUS NOT BE EQUAL TO THE COMMANDED STATE AN ALARM SHALL BE GENERATED AT THE CENTRAL WORKSTATION. ADDITIONALLY A CHILLER FAILURE ALRM WILL ALSO BE GENERATED AT THE CENTRAL WORKSTATION.

NOTES:

(E) EXISTING DEVICE.

(M) DEVICE PROVIDED BY MANUFACTURER.

CHILLED WATER SYSTEM

LOCATED AT PENTHOUSE FLOOR.
REFER TO SHEET 9 FOR PANEL TERMINATIONS.

"AS BUILT"

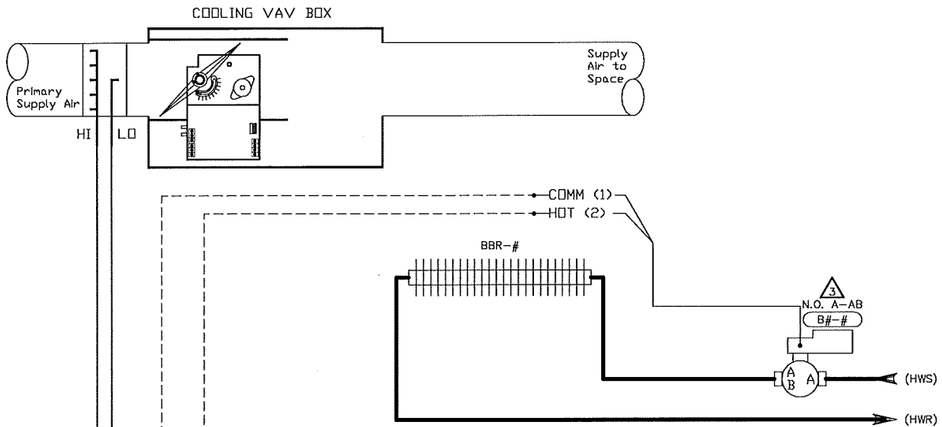
LONG
Denver 303-975-2100
Colorado Springs 719-592-0364
Wyoming 307-265-5997

CITY OF COLORADO SPRINGS - EPC (CITY)
CHILLED WATER SYSTEM CONTROL DIAGRAM
30 SOUTH NEVADA AVENUE
COLORADO SPRINGS, COLORADO 80901

NO.	Revision:	Date:	By:
A	ENGINEERING REVIEW	10-12-07	BD
B	TC2955 AS BUILT	08-20-08	CJJ
C	AS BUILT	11-07-08	JJV

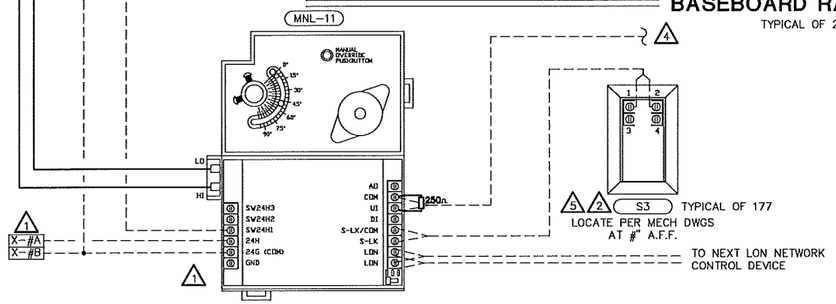
Engineer: BURNS & MCDONNELL
Contractor: BURNS & MCDONNELL
Proj Mgr.: B. STUBER
App. Eng.: B. DHAKAL
Created: 08-22-07
Standard Template: N/A

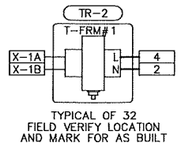
JOB NUMBER TC2955
SHEET NO. 10



BASEBOARD RADIATION

TYPICAL OF 26





TYPICAL OF 32
FIELD VERIFY LOCATION AND MARK FOR AS BUILT

VAV BOX COOLING ONLY

TYPICAL OF 180

A TYPICAL OF 1 ADDED FOR TC3082

BILL OF MATERIAL

Tag	Qty	Part #	Supplier	Description
MNL-11	180	MNL-V2RV3	TAC	LONMARK FAN-POWERED VAV W/REHEAT CONTROLLER
S3	177	MN-S3	TAC	IA SENSOR W/ STPT & OVRD
TR-2	32	T-208	TAC	96VA 120/24VAC TRANSFORMER

A TC3082 BILL OF MATERIAL

Tag	Qty	Part #	Supplier	Description
MNL-11	1	MNL-V2RV3	TAC	LONMARK FAN-POWERED VAV W/REHEAT CONTROLLER
S3	1	MN-S3	TAC	IA SENSOR W/ STPT & OVRD

SEQUENCE OF OPERATION

THE TAC MNL-V2RV3 I/A SERIES DIRECT DIGITAL CONTROLLER (DDC) PROVIDES CONTROL OF THE TERMINAL UNIT AS FOLLOWS.

A USER ADJUSTABLE OCCUPIED/UNOCCUPIED TIME SCHEDULE CONTROLS THE MODE OF OPERATION IN CONJUNCTION WITH THE AIR HANDLING UNIT.

VAV BOX:
OCCUPIED MODE:
THE DDC MODULATES THE PRIMARY AIR DAMPER FROM THE MINIMUM CFM SETTING TO THE MAXIMUM CFM SETTING TO MAINTAIN THE SPACE TEMPERATURE (S3) AT THE OCCUPIED COOLING SET POINT INITIALLY SET AT 74 DEGF(ADJ.). THE SPACE TEMPERATURE SETPOINT RANGE IS LIMITED IN ADJUSTMENT TO A SPAN OF +/- 2 DEGF. (ADJ) IF THE ZONE TEMPERATURE FALLS BELOW THE OCCUPIED HEATING SET POINT INITIALY SET AT 70 DEGF (ADJ.) UPON A FURTHER DROP IN ZONE TEMPERATURE, THE BASEBOARD ZONE VALVE IS OPENED AND THE VALUE OF THE DEVIATION BELOW THE HEATING SETPOINT IS CALCULATED AT THE UNC CONTROLLER AGAINST THE TOTAL NUMBER OF ZONES AND APPLIED AS A WEIGHTED VALUE TO PROVIDE A RESET SET POINT OF THE SECONDARY HOT WATER LOOPS SERVING THE ASSOCIATED BASEBOARD ZONE TO A MAXIMUM TEMPERATURE EQUAL TO THE PRIMARY LOOP RESET SET POINT. (SEE MATRIX FOR SPECIFIC VAV BOXES AND THEIR ASSOCIATED BASEBOARD ZONES AND CONTROLLER ### FOR SEQUENCE OF OPERATION FOR SECONDARY LOOPS)

UNOCCUPIED MODE
THE VAV AND AIR HANDLING SYSTEM IS NOT UTILIZED FOR UNOCCUPIED HEATING OR COOLING. THE VAV AIR DAMPER IS POSITIONED CLOSED DURING THE UNOCCUPIED MODE TO PREVENT BACKDRAFTING THE AIR HANDLING UNIT.
UNOCCUPIED HEATING IS ACCOMPLISHED UTILIZING THE BASEBOARD HEATING ONLY.

MORNING WARM-UP:
THE CONTROLLERS ACTION IS REVERSED BASED ON DISCHARGE AIR TEMPERATURE OF THE AIR HANDLING UNIT. INITIALLY SET AT 80 DEGF (ADJ) TO ENTER THE REVERSE ACTION MODE AND 75 DEGF (ADJ) TO RETURN TO THE DIRECT ACTING MODE. THE PRIMARY AIR DAMPER IS MODULATED FROM THE MAXIMUM CFM SET POINT TO THE MINIMUM CFM SET POINT TO MAINTAIN THE ZONE TEMPERATURE AT THE OCCUPIED HEATING SET POINT (ADJ.). IF ANY ZONE IS BELOW THE OCCUPIED HEATING SET POINT, THE DDC ENABLES THE AHU IN THE MORNING WARM-UP MODE (SEE AHU SEQUENCE). ONCE THE COLDEST ZONE REACHES THE OCCUPIED HEATING SET POINT, THE UNIT REVERTS TO THE OCCUPIED MODE.

NOTES:

1. VAV CONTROLLER POWER CONSUMPTION IS 12VA AND BASEBOARD VALVE IS 1 VA. USE A MAXIMUM OF 77VA (APROX. 80%) FROM ONE 96 VA TRANSFORMER.
2. SELECTED SENSORS TO HAVE LON COMMUNICATION. REFER TO SYSTEM ARCHITECTURE SHEET 4 FOR LOCATIONS AND DETAIL (R) ON SHEET 5 FOR WIRING.
3. REFER TO VALVE SCHEDULE ON SHEET 3.
4. SELECTED BOXES TO MONITOR RA-CO2 LEVEL. REFER TO SHEETS 6 AND 12 FOR DETAILS.

"AS BUILT"

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Wyoming 307-265-5997

CITY OF COLORADO SPRINGS - EPC (CITY)

VAV BOX CONTROL DIAGRAM
30 SOUTH NEVADA AVENUE
COLORADO SPRINGS, COLORADO 80901

NO.	Revision:	Date:	By:
A	ENGINEERING REVIEW	10-12-07	BD
B	TC3082 AS BUILT	03-28-08	JJV
C	TC2955 AS BUILT	06-20-08	CLJ
D			
E			
F			

Engineer: BURNS & MCDONNELL
Contractor: BURNS & MCDONNELL
Proj. Mgr.: B. STUBER
App. Eng.: B. DHAKAL
Created: 08-22-07
Standard Template: INV_LON_VAV_VAV_O.dwt

JOB NUMBER TC2955
SHEET NO. 11

VAV SCHEDULE

PLAN TAG #	SIZE	CLG CFM MAX	CLG CFM MIN	HTG CFM	POWER SOURCE	ADD'L DEVICE	ASSOC. RTU	PFP=PARALLEL SFP=SERIES VAV=PINCHDOWN	HW=HOT WATER ELEC#-ELECTRIC (# OF STAGES)
VAVB-01	10	900	405				AHU-1	VAV	
VAVB-02	8	650	195				AHU-1	VAV	
VAVB-03	8	800	272				AHU-1	VAV	
VAVB-04	8	500	150				AHU-1	VAV	
VAVB-05	10	1200	360				AHU-1	VAV	
VAVB-06	8	500	155				AHU-1	VAV	
VAVB-07	8	500	275				AHU-1	VAV	
VAVB-08	6	300	132				AHU-1	VAV	
VAVB-09	6	400	160			RA-CO2 SENSOR	AHU-1	VAV	
VAVB-10	12	1400	420				AHU-1	VAV	
VAVB-11	10	1050	315				AHU-1	VAV	
VAVB-12	10	1100	362				AHU-1	VAV	
VAVB-13	6	375	113				AHU-1	VAV	
VAVB-14	8	475	171				AHU-1	VAV	
VAV1-01	8	475	143			BB-# (WEST)	AHU-1	VAV	
VAV1-02	6	400	120				AHU-1	VAV	
VAV1-03	8	575	179				AHU-1	VAV	
VAV1-04	4	150	60				AHU-1	VAV	
VAV1-05	8	475	157				AHU-1	VAV	
VAV1-06	4	230	69				AHU-1	VAV	
VAV1-07	N/A	1975	593			STAT FOR VAV-1-8	AHU-1	VAV	
VAV1-08	N/A	1275	383				AHU-1	VAV	
VAV1-09	8	725	219				AHU-1	VAV	
VAV1-10	5	225	77				AHU-1	VAV	
VAV1-11	6	325	117				AHU-1	VAV	
VAV1-12	8	800	240				AHU-1	VAV	
VAV1-13	5	250	85				AHU-1	VAV	
VAV1-14	6	375	138				AHU-1	VAV	
VAV1-15	8	800	240				AHU-1	VAV	
VAV1-16	8	550	223				AHU-1	VAV	
VAV1-17	8	700	210				AHU-1	VAV	
VAV1-18	8	575	208				AHU-1	VAV	
VAV1-19	8	500	150			RA-CO2 SENSOR	AHU-1	VAV	
VAV1-20	8	800	240			BB-# (SOUTH)	AHU-1	VAV	
VAV1-21	10	1000	300				AHU-1	VAV	
VAV1-22	8	800	240			BB-# (SOUTHEAST)	AHU-1	VAV	
VAV1-23	6	440	132				AHU-1	VAV	
VV-1	N/A	N/A	N/A				AHU-1	VAV	
VV-2	N/A	N/A	N/A			STAT FOR VV1-4	AHU-1	VAV	
VV-3	N/A	N/A	N/A				AHU-1	VAV	
VV-4	N/A	N/A	N/A				AHU-1	VAV	
VAV2-01	6	400	120			BB-# WEST	AHU-1	VAV	
VAV2-02	8	700	210				AHU-1	VAV	
VAV2-03	8	700	210				AHU-1	VAV	
VAV2-04	6	400	120				AHU-1	VAV	
VAV2-05	8	700	210				AHU-1	VAV	
VAV2-06	8	650	195				AHU-1	VAV	
VAV2-07	8	920	276				AHU-1	VAV	
VAV2-08	10	1200	424				AHU-1	VAV	
VAV2-09	8	800	240				AHU-1	VAV	
VAV2-10	5	300	90				AHU-1	VAV	
VAV2-11	8	700	230				AHU-1	VAV	
VAV2-12	10	1000	300				AHU-1	VAV	
VAV2-13	8	800	240				AHU-1	VAV	
VAV2-14	8	600	180			BB-# (WEST)	AHU-1	VAV	
VAV2-15	8	400	124				AHU-1	VAV	
VAV2-16	8	925	278				AHU-1	VAV	
VAV2-17	8	775	237				AHU-1	VAV	
VAV2-18	8	800	240				AHU-1	VAV	
VAV2-19	8	700	302			RA-CO2 SENSOR	AHU-1	VAV	
VAV2-20	10	1300	390			BB-# (SOUTH)	AHU-1	VAV	
VAV2-21	5	360	108				AHU-1	VAV	
VAV2-22	6	520	156			BB-# (SOUTHEAST)	AHU-1	VAV	
VAV2-23	8	450	144				AHU-1	VAV	
VAV3-01	8	475	143			BB-# (NORTH)	AHU-1	VAV	
VAV3-02	8	500	150				AHU-1	VAV	
VAV3-03	4	200	72				AHU-1	VAV	
VAV3-04	5	350	105				AHU-1	VAV	
VAV3-05	8	925	278				AHU-1	VAV	
VAV3-06	8	600	180				AHU-1	VAV	
VAV3-07	6	450	135				AHU-1	VAV	
VAV3-08	8	600	180				AHU-1	VAV	
VAV3-09	8	725	236				AHU-1	VAV	
VAV3-10	10	900	270				AHU-1	VAV	
VAV3-11	10	950	285			BB-# (WEST)	AHU-1	VAV	
VAV3-12	6	400	138				AHU-1	VAV	
VAV3-13	8	700	231				AHU-1	VAV	
VAV3-14	8	925	278				AHU-1	VAV	
VAV3-15	6	400	120			BB-# (SOUTH)	AHU-1	VAV	
VAV3-16	10	1250	375				AHU-1	VAV	
VAV3-17	6	425	153			RA-CO2 SENSOR	AHU-1	VAV	
VAV3-18	6	400	120			BB-# (SOUTHEAST)	AHU-1	VAV	
VAV3-19	5	300	90				AHU-1	VAV	
VAV3-20	6	520	200				AHU-1	VAV	
VAV3-21	10	1375	413				AHU-1	VAV	
VAV3-22	6	350	112				AHU-1	VAV	
VAV4-01	5	300	90				AHU-1	VAV	
VAV4-02	10	900	270				AHU-1	VAV	
VAV4-03	8	600	180			BB-# (NORTH)	AHU-1	VAV	
VAV4-04	8	900	270				AHU-1	VAV	
VAV4-05	8	500	150				AHU-1	VAV	
VAV4-06	8	800	240				AHU-1	VAV	
VAV4-07	8	800	262				AHU-1	VAV	
VAV4-08	8	800	240				AHU-1	VAV	
VAV4-09	8	700	210				AHU-1	VAV	
VAV4-10	5	300	90				AHU-1	VAV	
VAV4-11	8	700	210			BB-# (WEST)	AHU-1	VAV	
VAV4-12	10	850	255				AHU-1	VAV	
VAV4-13	8	925	278				AHU-1	VAV	
VAV4-14	8	600	180			BB-# (SOUTH)	AHU-1	VAV	
VAV4-15	8	600	180			RA-CO2 SENSOR	AHU-1	VAV	
VAV4-16	10	1400	420				AHU-1	VAV	
VAV4-17	8	700	210				AHU-1	VAV	
VAV4-18	6	520	156			BB-# (SOUTHEAST)	AHU-1	VAV	
VAV4-19	6	520	156			BB-# (SOUTHEAST)	AHU-1	VAV	
VAV4-20	8	925	278				AHU-1	VAV	
VAV4-21	8	800	240				AHU-1	VAV	
VAV4-22	10	1275	350				AHU-1	VAV	
VAV4-23	6	350	105				AHU-1	VAV	
VAV5-01	10	1000	300				AHU-1	VAV	
VAV5-02	8	650	195			BB-# (NORTH)	AHU-1	VAV	
VAV5-03	8	800	240				AHU-1	VAV	
VAV5-04	8	600	180				AHU-1	VAV	
VAV5-05	8	750	225				AHU-1	VAV	
VAV5-06	6	400	140				AHU-1	VAV	
VAV5-07	10	1075	323				AHU-1	VAV	
VAV5-08	10	1050	315				AHU-1	VAV	
VAV5-09	5	300	90				AHU-1	VAV	
VAV5-10	10	1000	300			BB-# (WEST)	AHU-1	VAV	
VAV5-11	8	750	229				AHU-1	VAV	
VAV5-12	6	400	120				AHU-1	VAV	
VAV5-13	10	975	293				AHU-1	VAV	
VAV5-14	8	600	180			BB-# (SOUTH)	AHU-1	VAV	
VAV5-15	8	600	182				AHU-1	VAV	
VAV5-16	12	1700	510				AHU-1	VAV	
VAV5-17	6	400	120			BB-# (SOUTHEAST)	AHU-1	VAV	
VAV5-18	8	600	212				AHU-1	VAV	
VAV5-19	5	250	75			RA-CO2 SENSOR	AHU-1	VAV	
VAV5-20	8	925	278				AHU-1	VAV	
VAV5-21	10	1050	315				AHU-1	VAV	
VAV5-22	8	675	223				AHU-1	VAV	
VAV5-23	6	350	112				AHU-1	VAV	
VAV6-01	10	1100	310				AHU-1	VAV	
VAV6-02	8	700	210			BB-# (NORTH)	AHU-1	VAV	
VAV6-03	8	925	278				AHU-1	VAV	
VAV6-04	10	975	293				AHU-1	VAV	
VAV6-05	6	450	135				AHU-1	VAV	
VAV6-06	10	1000	300				AHU-1	VAV	
VAV6-07	5	300	90				AHU-1	VAV	
VAV6-08	8	700	210				AHU-1	VAV	
VAV6-09	10	875	283				AHU-1	VAV	
VAV6-10	6	400	120				AHU-1	VAV	
VAV6-11	10	950	285			BB-# (WEST)	AHU-1	VAV	
VAV6-12	10	900	270				AHU-1	VAV	
VAV6-13	10	1100	330				AHU-1	VAV	
VAV6-14	8	600	180			BB-# (SOUTH)	AHU-1	VAV	
VAV6-15	10	1100	330				AHU-1	VAV	
VAV6-16	8	750	225				AHU-1	VAV	
VAV6-17	6	400	120			BB-# (SOUTHEAST)	AHU-1	VAV	
VAV6-18	6	400	172				AHU-1	VAV	
VAV6-19	8	800	240				AHU-1	VAV	
VAV6-20	6	400	120			RA-CO2 SENSOR	AHU-1	VAV	
VAV6-21	10	1150	345				AHU-1	VAV	
VAV6-22	10	1025	308				AHU-1	VAV	
VAV6-23	6	450	167				AHU-1	VAV	
VAV6-24	6	350	105				AHU-1	VAV	
VAV6-25	8	400	120				AHU-1	VAV	
VAV7-01	10	900	270				AHU-1	VAV	
VAV7-02	8	500	150			BB-# (NORTH)	AHU-1	VAV	
VAV7-03	10	950	285				AHU-1	VAV	
VAV7-04	8	500	150				AHU-1	VAV	
VAV7-05	5	250	90				AHU-1	VAV	
VAV7-06	5	300	108				AHU-1	VAV	
VAV7-07	8	700	224				AHU-1	VAV	
VAV7-08	6	400	120				AHU-1	VAV	
VAV7-09	5	300	90				AHU-1	VAV	
VAV7-10	10	900	270				AHU-1	VAV	
VAV7-11	10	950	285			BB-# (WEST)	AHU-1	VAV	
VAV7-12	4	200	60				AHU-1	VAV	
VAV7-13	10	875	294				AHU-1	VAV	
VAV7-14	12	1300	390				AHU-1	VAV	
VAV7-15	8	700	210			RA-CO2 SENSOR	AHU-1	VAV	
VAV7-16	6	500	150			BB-# (SOUTH)	AHU-1	VAV	
VAV7-17	12	1500	450				AHU-1	VAV	
VAV7-18	8	650	195				AHU-1	VAV	
VAV7-19	8	600	180			BB-# (SOUTHEAST)	AHU-1	VAV	
VAV7-20	10	900	270				AHU-1	VAV	
VAV7-21	6	325	98				AHU-1	VAV	
VAV7-22	10	1200	360				AHU-1	VAV	
VAV7-23	8	500	161				AHU-1	VAV	
VAV7-24	6	350	105				AHU-1	VAV	

FLOOR	BASE BOARD	VAV BOX
1	BB-WEST	VAV1-1, VAV1-12, VAV1-15, VAV1-17, VAV1-20
1	BB-SOUTH	VAV1-20, VAV1-21
1	BB-SOUTH EAST	VAV1-6, VAV1-19, VAV1-22
2	BB-WEST	VAV2-14, VAV2-16, VAV2-18
2	BB-SOUTH	VAV2-18, VAV2-20, VAV2-22
2	BB-SOUTH EAST	VAV2-4, VAV2-5, VAV2-6, VAV2-7, VAV2-21, VAV2-22
3	BB-WEST	VAV3-11, VAV3-14, VAV3-15
3	BB-SOUTH	VAV3-15, VAV3-16, VAV3-18
3	BB-SOUTH EAST	VAV3-04, VAV3-05, VAV3-06, VAV3-18, VAV3-19, VAV3-20, VAV3-21
3	BB-NORTH	VAV3-01, VAV3-02, VAV3-03, VAV3-04, VAV3-10, VAV3-11
4	BB-WEST	VAV4-11, VAV4-13, VAV4-14
4	BB-SOUTH	VAV4-14, VAV4-16, VAV4-18
4	BB-SOUTH EAST	VAV4-03, VAV4-04, VAV4-05, VAV4-18, VAV4-19, VAV4-20, VAV4-22
4	BB-NORTH	VAV4-01, VAV4-02, VAV4-03, VAV4-09, VAV4-10, VAV4-11
5	BB-WEST	VAV5-12, VAV5-13, VAV5-14
5	BB-SOUTH	VAV5-14, VAV5-16, VAV5-17
5	BB-SOUTH EAST	VAV5-02, VAV5-03, VAV5-04, VAV5-17, VAV5-20, VAV5-21
5	BB-NORTH	VAV5-01, VAV5-02, VAV5-08, VAV5-10, VAV5-12
6	BB-WEST	VAV6-11, VAV6-13, VAV6-14
6	BB-SOUTH	VAV6-14, VAV6-15, VAV6-17
6	BB-SOUTH EAST	VAV6-02, VAV6-03, VAV6-04, VAV6-17, VAV6-18, VAV6-21, VAV6-22
6	BB-NORTH	VAV6-01, VAV6-02, VAV6-08, VAV6-10, VAV6-11
7	BB-WEST	VAV7-11, VAV7-14, VAV7-16
7	BB-SOUTH	VAV7-16, VAV7-17, VAV7-19
7	BB-SOUTH EAST	VAV7-02, VAV7-03, VAV7-04, VAV7-19, VAV7-20, VAV7-22
7	BB-NORTH	VAV7-01, VAV7-02, VAV7-10, VAV7-11

"AS BUILT"

LONG
Denver 303-975-2100
Colorado Springs 719-592-0364
Wyoming 307-265-5997

CITY OF COLORADO SPRINGS - EPC (CITY)

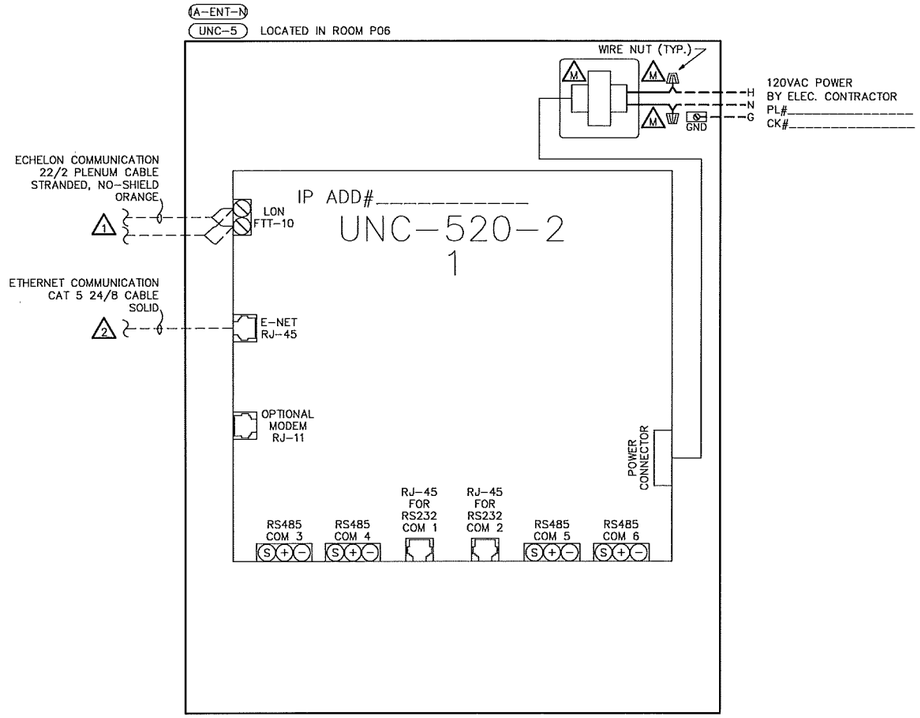
VAV TABLE
30 SOUTH NEVADA AVENUE
COLORADO SPRINGS, COLORADO 80901

NO.	Revision:	Date:	By:
A	ENGINEERING REVIEW	10-12-07	BD
B	TC3082 AS BUILT	03-28-08	JJV
C	TC2955 AS BUILT	06-20-08	CLV
D	AS BUILT	11-07-08	JJV

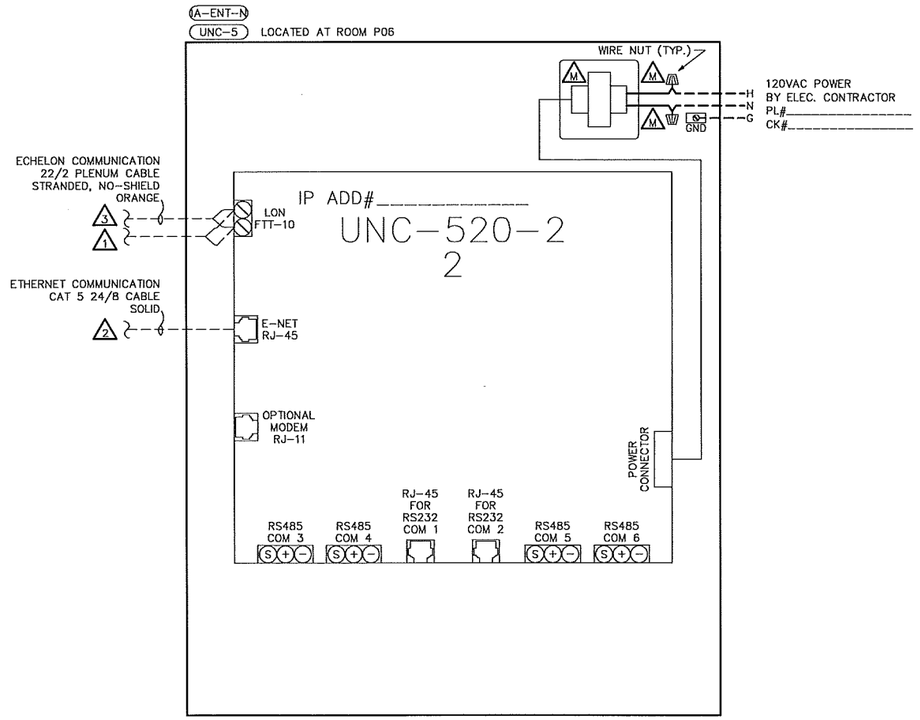
Engineer: BURNS & MCDONNELL
Contractor: BURNS & MCDONNELL
Proj Mgr.: B. STUBER
App. Engr.: B. DHAKAL
Created: 08-22-07

Standard Template: N/A

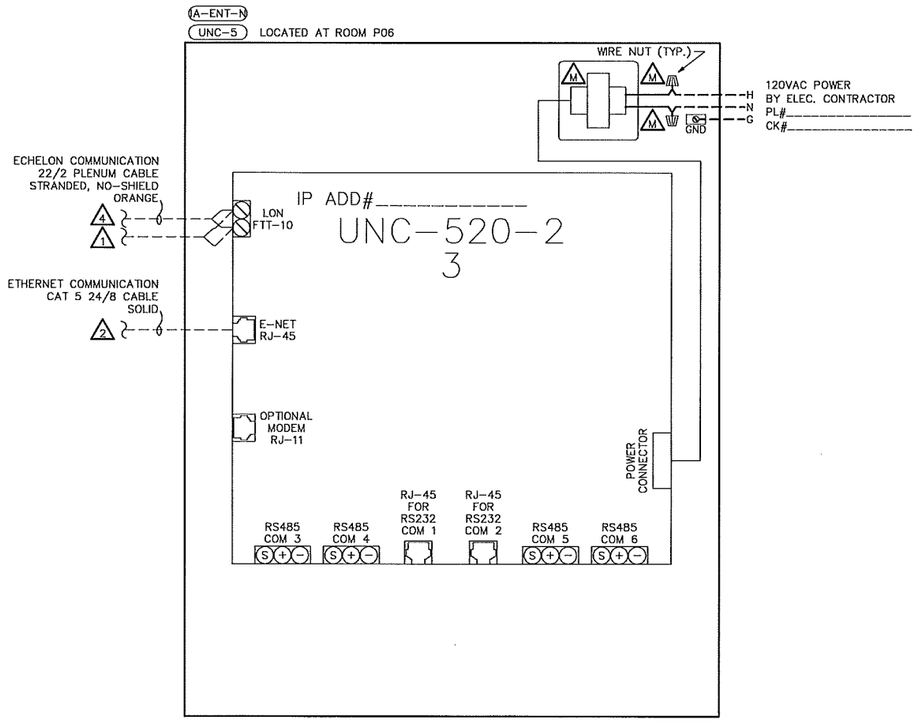
JOB NUMBER TC2955
SHEET NO. 12



TCP-3 NETWORK CONTROL PANEL



TCP-4 NETWORK CONTROL PANEL



TCP-5 NETWORK CONTROL PANEL

BILL OF MATERIAL

Tag	Qty	Part #	Supplier	Description
IA-ENT-N	3	IA-ENT-N	TAC	ENTERPRISE SERVER LICENSE FOR ADDITIONAL UNC
UNC-5	3	UNC-520-2	TAC	NETWORK CONTROLLER

NOTES:

1. LON COMM. TRUNK (SEE SHEET 4).
2. TO ETHERNET (SEE SHEET 4).
3. TO BOILER LON COMMUNICATION CARD
4. TO CHILLER LON COMMUNICATION CARD
M. DEVICE PROVIDED BY MANUFACTURER.

"AS BUILT"

LONG
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Colorado Springs 719-592-4364
Wyoming 307-265-5997

CITY OF COLORADO SPRINGS - EPC (CITY)
UNC NETWORK CONTROLLERS
30 SOUTH NEVADA AVENUE
COLORADO SPRINGS, COLORADO 80901

NO.	Revision:	Date:	By:
A	TC2955 AS BUILT	06-20-08	CLJ
B			
C			
D			
E			
F			

Engineer: BURNS & MCDONNELL
Contractor: BURNS & MCDONNELL
Proj. Mgr.: B. STUBER
App. Eng.: B. DHAKAL
Created: 08-22-07
Standard Template: N/A

JOB NUMBER TC2955
SHEET NO. 13